

Inference on Welfare and Value Functionals under Optimal Treatment Assignment

Xiaohong Chen,^{*} Zhenxiao Chen,[†] and Wayne Yuan Gao[‡]

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Abstract

We study estimation and inference for a class of welfare and value functionals of the nonparametric conditional average treatment effect (CATE) under optimal treatment assignment. For the optimal welfare functional, we establish $\sqrt{n}$ asymptotic normality of semiparametric plug-in estimators with an analytical variance formula. For more general value functionals, the plug-in estimators are instead asymptotically normal at the slower one-dimensional nonparametric rate $n^{-\frac{s}{2s+1}}$, where s is the Hölder smoothness of the CATE, with a consistent sieve variance estimator; we show this rate is minimax optimal, and develop split-sample and leave-one-out debiased estimators that attain it under weaker smoothness conditions. Building on the sieve Riesz representer, we further develop a unified debiased machine learning procedure that accommodates generic machine-learning first stages and applies whether the functional is regular (yielding $\sqrt{n}$ inference) or irregular (yielding the slower minimax rate), nesting the welfare and value functionals as special cases. Monte Carlo simulations document good finite-sample performance, and we apply our methods to study the effectiveness of job training programs on earnings using the JTPA data set.

Keywords: optimal treatment assignment, conditional average treatment effect, semiparametric estimation and inference, regular and irregular functionals, minimax optimality, debiased estimation

^{*}Department of Economics and Cowles Foundation for Economic Research, Yale University, 28 Hillhouse Ave, New Haven, CT 06511, USA, xiaohong.chen@yale.edu

[†]Department of Economics, University of Pennsylvania, 133 S 36th St., Philadelphia, PA 19104, USA, zxchen@upenn.edu.

[‡]Department of Economics, University of Pennsylvania, 133 S 36th St., Philadelphia, PA 19104, USA, waynegao@upenn.edu.

1 Introduction

In this paper, we study the estimation and inference on welfare and value functionals of a given treatment under optimal (“first-best”) treatment assignment.

Let $D_i \in \{0, 1\}$ denote a certain binary treatment for subject i , $(Y_i(0), Y_i(1))$ denote the potential outcomes of interest, and $Y_i := Y_i(D_i) \in \mathbb{R}$ denote the observed outcome. Let $X_i \in \mathbb{R}^{d_x}$ denote subject i ’s observable characteristics, which are distributed with density f_0 . We suppose that researchers have access to a random sample of training data $\{(D_i, Y_i, X_i)\}_{i=1}^n$.

Under the standard conditional unconfoundedness assumption $(Y_i(0), Y_i(1)) \perp D_i | X_i$ and the overlap condition $p_0(x) := \mathbb{E}[D_i | X_i = x] \in (0, 1)$, the conditional average treatment effect (CATE) defined by

$$\text{CATE}(x) := \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x]$$

is identified from data by

$$\text{CATE}(x) \equiv h_0(x) := \mu_0(x, 1) - \mu_0(x, 0), \quad (1)$$

where $h_0 : \mathbb{R}^{d_x} \mapsto \mathbb{R}$, and

$$\mu_0(x, d) := \mathbb{E}[Y_i | X_i = x, D_i = d] \quad (2)$$

is the nonparametric regression function of the outcome Y_i on X_i and D_i . We maintain the conditional unconfoundedness assumption and the overlap condition, and will thereafter simply refer to h_0 as the CATE function. We further assume that h_0 belongs to a Hölder class of functions with smoothness $s > 1$.

We consider a standard scenario where policymakers can assign treatments based on covariates, and focus on the following two core types of welfare and value parameters. The first type is the maximized welfare of the target population under optimal treatment assignment:

$$W(h_0) := \int [h_0(x)]_+ f(x) dx \quad (3)$$

where f is the marginal density of x in the target population and $[t]_+ := \max(t, 0)$ is the rectified linear unit (ReLU) function. $W(h_0)$ averages the CATE over the population under the “first-best” treatment assignment rule: “treat type x if and only if $\text{CATE}(x) \geq 0$,” and is thus often referred to as the welfare under optimal treatment assignment. We will thereafter refer to $W(h_0)$ as the welfare functional.

Alternatively, one may also be interested in evaluating the average of a value other than CATE over the population under optimal treatment assignment:

$$V(h_0) := \int \mathbb{1}\{h_0(x) \geq 0\} v_0(x) f(x) dx \quad (4)$$

where $v_0 : \mathbb{R}^{d_x} \mapsto \mathbb{R}$ is a user-defined function that may be a utility function, a cost function, or any economically meaningful function of the observed covariate x . For example, setting $v_0(x) = a'x$ endows $V(h_0)$ with the interpretation as certain aggregate characteristics of the treated population under optimal treatment assignment, and setting $v_0 \equiv 1$ implies that $V(h_0) \equiv \mathbb{P}_f(h_0(X_i) \geq 0)$ becomes the share of the target population to be treated (i.e. with nonnegative CATE).

In the formulation of $W(h_0)$ and $V(h_0)$ above, we take the density $f(x)$ to be known. This is in itself relevant in settings where the covariate density of the target population is configured or known/estimated from other sources than the training sample used to estimate CATE. For example, CATE may be estimated from a smaller pilot program, while the policymakers are contemplating implementing the policy on a statewide or nationwide basis with a much larger population. That said, in this paper we also consider an important case where f is unknown and set to be the covariate density in the underlying population of the training sample. In this case, f does not need to be estimated, as the integral with respect to f can be naturally approximated via sample average in the training sample. This corresponds more closely to the “empirical welfare” as considered in [Kitagawa and Tetenov \(2018\)](#). We provide results for this setting as well.

In this paper, we establish estimation and inference results for the welfare and value functionals $W(h_0)$ and $V(h_0)$ with nonparametrically estimated CATE h_0 . Our main theoretical results can be summarized informally as follows.

First, for the welfare functional $W(h_0)$, we establish that semiparametric plug-in estimators are asymptotically normal at the parametric $\sqrt{n}$ rate and derive closed-form asymptotic variance formulas, along with consistent asymptotic variance estimators. The key insight of the $\sqrt{n}$ rate is geometric: by the definition of the welfare functional, the integrand $h_0(x)$ vanishes on the boundary of the region of integration $\{x : h_0(x) = 0\}$, neutralizing the non-smoothness of the indicator function $\mathbb{1}\{h_0(x) \geq 0\}$. Because $W(h_0)$ is regular, it further admits a Neyman-orthogonal (locally robust) moment; the resulting cross-fitted debiased machine learning estimator retains $\sqrt{n}$ -normality while accommodating generic machine-learning first-stage estimators and dispensing with Donsker-type conditions (Section 5.5).

Second, for the value functional $V(h_0)$ with general weight v_0 that does not vanish on the boundary $\{x : h_0(x) = 0\}$, we show that the rate of convergence is slower than $\sqrt{n}$, and is instead given by the 1-dimensional nonparametric regression rate $n^{-\frac{s}{2s+1}}$ under appropriate conditions. We establish asymptotic normality of the semiparametric plug-in estimator under this irregular convergence rate, and provide a consistent variance estimator based on the sieve Riesz representer. In particular, the consistent variance estimator features a Hausdorff

integral on the boundary submanifold $\{x : h_0(x) = 0\}$, for which we provide a numerical integration/differentiation method for computation.

Third, adapting the minimax lower bound theory in [Chen and Gao \(2026\)](#) to the treatment assignment setting, we show that the 1-dimensional nonparametric rate $n^{-\frac{s}{2s+1}}$ is the fastest possible convergence rate for the value functional among *all* estimators — including, in particular, estimators based on debiased machine learning or smoothing techniques. Combined with our asymptotic normality result, this establishes the minimax rate optimality of the semiparametric plug-in estimator, and clarifies that the slower-than- $\sqrt{n}$ rate is an intrinsic feature of the value functional under optimal treatment assignment rather than a limitation of the plug-in approach.

Fourth, we provide two types of debiased estimators for the value functional with improved properties relative to the simple plug-in estimator. Since the rate optimality of the plug-in estimator requires a relatively strong smoothness condition ($s \geq d_x$, with d_x the dimension of the covariates), we adapt the split-sample and leave-one-out debiased sieve estimators of [Chen and Gao \(2026\)](#) to the treatment assignment setting; these debiased estimators attain the minimax-optimal rate and asymptotic normality under the substantially weaker smoothness requirement $s > (d_x + 1)/2$. Furthermore, we develop a cross-fitted debiased estimator based on the sieve influence function (Section 5.6), which extends the double/debiased machine learning (DML) approach — originally designed for regular, $\sqrt{n}$ -estimable functionals — to the irregular value functional: the resulting estimator accommodates generic machine learning first-stage estimators of the CATE, with first-stage rate requirements that are in fact weaker than their regular DML counterparts.

We conduct an array of Monte Carlo experiments to document the good finite-sample accuracy of our theoretical inferential results. We show that the proposed standard error estimators perform well in finite sample, and the corresponding confidence intervals based on the asymptotic normality result and the standard error estimators have coverage probabilities close to their nominal levels. These findings hold not only for the $\sqrt{n}$ -estimable welfare functional, but also for the value functional, which is estimated at slower-than- $\sqrt{n}$ rate with standard errors computed through numerical integration and differentiation.

We also apply our results to empirical data from the Job Training Partnership Act (JTPA) data set. Following [Kitagawa and Tetenov \(2018\)](#), we take 30-month post-program earnings as the outcome variable and consider two covariates: pre-program earnings and education. We then provide empirical estimates and confidence intervals for two parameters: the welfare under first-best treatment assignment, which is $\sqrt{n}$ -estimable, and the share of population to be treated under first-best treatment assignment, which is not $\sqrt{n}$ -estimable. As in [Kitagawa and Tetenov \(2018\)](#), we also consider two different scenarios: one with the cost of the

treatment incorporated, and one without. These parameters have also been estimated in Kitagawa and Tetenov (2018) under the label of “nonparametric plug-in rule” using kernel first-stages, but Kitagawa and Tetenov (2018) only provide point estimates with no confidence intervals for them. We use sieve (B-spline) first-stage nonparametric estimators and find similar results to those in Kitagawa and Tetenov (2018), and further provide informative confidence intervals for both the welfare and the share parameters.

Closely Related Literature

This work is a companion paper to Chen and Gao (2026), and directly applies the general theoretical results there to handle differentiation with respect to region of integration and semiparametric estimation of integrals over submanifolds. Specifically, this work focuses on the important context of treatment assignment problems, and deals with two features that are not discussed in Chen and Gao (2026). First, CATE is defined as the difference of two nonparametric regression functions between the treated and untreated subpopulation (or the difference of two point evaluations of a nonparametric function with treatment status defined as an argument too), and is often estimated as the difference of two first-stage nonparametric estimators. Second, in treatment assignment problems, researchers may face two different scenarios in terms of the distribution of the covariates: sometimes, this distribution can be treated as known and may be different from the covariate distribution in the experimental/observational population from which CATE is estimated; at other times, one may want to treat the covariate distribution as unknown and the same as the experimental/observational population, and use sample averages to automatically incorporate the covariate distribution. In this paper, we take into account these special structures of the problem, and establish the inference results by providing lower-level sufficient conditions for the general theory in Chen and Gao (2026). In addition, we adapt the minimax lower bound results and the split-sample and leave-one-out debiased estimators in Chen and Gao (2026) to the treatment assignment setting, thereby establishing the minimax rate optimality of our estimation and inference procedures for the value functional.

To delineate what is new here relative to Chen and Gao (2026), it is useful to separate three kinds of results. First, *new theorems and proofs* specific to the treatment-assignment problem: the $\sqrt{n}$ -asymptotic normality and closed-form variance of the welfare functional under known and unknown target density (Theorems 1–2); a unified debiased machine learning procedure built on the sieve Riesz representer, valid whether the functional is regular or irregular (Theorem 7) and specialized to the welfare and value functionals (Corollaries 1 and 2); and the unknown-density “empirical welfare” result for the value functional (Theorem 4). Second, *specializations of companion results* to the CATE setting, obtained by

mapping the general submanifold-integral theory of [Chen and Gao \(2026\)](#) to the weight $w = v_0 f$, submanifold dimension $m = d_x - 1$, and the difference-of-regressions structure $h_0 = \mu_0(\cdot, 1) - \mu_0(\cdot, 0)$: the plug-in rate and inference for the value functional (Theorem 3), the minimax lower bound (Theorem 5), and the split-sample and leave-one-out debiased estimators (Theorem 6). Third, *computational adaptations* required by the binary-treatment random design: the stacked sieve regression (17) for the difference of arm-specific regressions, and the numerical integration/differentiation scheme for the boundary (Hausdorff) variance integral.

Our paper makes new contributions to the literature on estimation and inference on a general value functional of a policy under optimal treatment assignment. To the best of our knowledge, all the existing work on limiting distributions of functionals of a policy under optimal treatment assignments considered $\sqrt{n}$ -normality only. Our paper is the first to establish slower-than- $\sqrt{n}$ limiting distribution and inference results for irregular value functionals of a policy under optimal treatment assignment, along with a matching minimax lower bound showing that the slower-than- $\sqrt{n}$ rate cannot be improved upon by any estimator. Previously, [Bhattacharya and Dupas \(2012\)](#) establishes $\sqrt{n}$ -normality of the optimal welfare value under budget constraint, using a Nadaraya-Watson kernel estimator for first-stage estimation of CATE. The important work of [Kitagawa and Tetenov \(2018\)](#) focuses on a related but slightly different topic: empirical welfare maximization within a constrained class of policy rules. That said, [Kitagawa and Tetenov \(2018\)](#) also considers and reports empirical estimates on the nonparametric plug-in rule, which can be interpreted as a plug-in estimator of the first-best welfare, though there were no theoretical results or confidence intervals for this estimate. The recent papers by [Park \(2025\)](#) and [Whitehouse, Austern and Syrgkanis \(2025\)](#) use soft-max functions to smooth the max function, and apply the debiased machine learning approach to establish asymptotic normality and provide inferential results, while [Whitehouse, Austern and Syrgkanis \(2025\)](#) establishes $\sqrt{n}$ asymptotic normality of their soft-max welfare functional. [Feng, Hong and Nekipelov \(2025\)](#) analyzes binary treatment assignment under constraints and the asymptotic property of the welfare of the policy under optimal cutoff choice, and establishes $\sqrt{n}$ asymptotic normality of the functionals.¹ Our paper complements the existing work in several ways. First, we clarify that the $\sqrt{n}$ -estimability of the welfare functional is due to the fact that the integrand (CATE) by construction vanishes on the boundary of the optimally treated population $\{x : \text{CATE}(x) = 0\}$, which is specific to the welfare functional but generally not satisfied for other types of value functionals. Second, we demonstrate that the $\sqrt{n}$ -estimability of the welfare functional can be

¹Technically, [Feng, Hong and Nekipelov \(2025\)](#), [Chen and Gao \(2026\)](#) and our paper all utilize differential geometry and Hausdorff (surface) integrals.

attained without the use of smoothing/soft-max functions. Third and most importantly, our results extend well beyond the welfare functional, and cover generic value functionals that may be slower than $\sqrt{n}$ -estimable as well — together with a matching minimax lower bound establishing that no estimator can do better than the slower-than- $\sqrt{n}$ rate for such value functionals. Fourth, while [Park \(2025\)](#) and [Whitehouse, Austern and Syrgkanis \(2025\)](#) apply the DML approach to smoothed (soft-max) versions of the welfare functional, which are regular functionals, our sieve-influence-function approach ([Section 5.6](#)) shows how the DML idea can be extended to the irregular, unsmoothed value functional itself, for which no influence function exists on the full Hilbert space.

The rest of the paper is organized as follows. [Section 2](#) provides a conceptual explanation of why the welfare functional $W(h_0)$ can be $\sqrt{n}$ -estimable while the value functional $V(h_0)$ is not in general. [Section 3](#) establishes the inference results for the welfare functional $W(h_0)$: semiparametric plug-in estimators under known and unknown covariate density, together with a variance estimator based on the sieve Riesz representer. [Section 4](#) provides corresponding results for the value functional $V(h_0)$, including the asymptotic normality of the plug-in estimator, a matching minimax lower bound, and split-sample and leave-one-out debiased estimators that attain the minimax-optimal rate under weaker smoothness conditions. [Section 5](#) develops a unified debiased machine learning procedure built on the sieve Riesz representer that accommodates generic machine-learning first-stage estimators of the CATE and applies whether the functional is regular or irregular, specializing to the welfare and value functionals. We report numerical results from Monte Carlo simulations in [Section 6](#), and conduct an empirical illustration in [Section 7](#). Proofs of theoretical results in the main text are available in [Appendix A](#).

2 Model and Functional Derivatives

We first introduce the standard treatment effect model, the general welfare functional of interest and the maintained assumptions in [Subsection 2.1](#). [Subsection 2.2](#) then derives the functional derivatives of the welfare and value functionals via the generalized Leibniz rule, and [Subsection 2.3](#) explains the key condition for $\sqrt{n}$ -estimability.

2.1 The Model and the Parameters of Interest

We first state the maintained assumptions of the paper. Let $\hat{h}(x) := \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$ be a nonparametric estimator of the CATE function, in which $\hat{\mu}(x, d)$ is a first-stage estimator

of the nonparametric regression model

$$Y_i = \mu_0(X_i, D_i) + \epsilon_i, \quad \mathbb{E}[\epsilon_i | X_i, D_i] = 0, \quad \mathbb{E}[\epsilon_i^2 | X_i, D_i] < \infty. \quad (5)$$

We impose the following basic assumptions in this paper.

Assumption 1 (Model). *The training data and the model satisfy:*

- (a) *Training sample: the training data $\{(Y_i, D_i, X_i)\}_{i=1}^n$ is a random sample drawn from $(Y, D, X) \in \mathbb{R} \times \{0, 1\} \times \mathcal{X}$ satisfying Model (5), where $\mathcal{X}$ is a bounded rectangular set (say $[0, 1]^{d_x}$) in $\mathbb{R}^{d_x}$, and X_i has a true unknown marginal density f_0 that is uniformly bounded away from zero and infinity on $\mathcal{X}$.*
- (b) *Strong overlap: $0 < \underline{p} \leq p_0(x) := \mathbb{E}[D_i | X_i = x] \leq \bar{p} < 1$ for some constants $0 < \underline{p} \leq \bar{p} < 1$.*
- (c) *Smoothness of regression function: For $d \in \{0, 1\}$, $\mu_0(\cdot, d) \in \Lambda^s(\mathcal{X})$ with $s > 1$.*

We now provide a heuristic overview of our theoretical analysis, and explain the key intuition for why the welfare functional $W(h_0)$ may be $\sqrt{n}$ -estimable (i.e., W is a regular functional) while $V(h_0)$ is generally not (i.e., V is an irregular functional).

We first introduce a more general value functional $\Phi(h_0)$ that nests $W(h_0)$ and $V(h_0)$ as special cases:

$$\Phi(h_0) := \int \mathbb{1}\{h_0(x) \geq 0\} \phi(h_0(x), x) f(x) dx, \quad (6)$$

where $\phi : \mathbb{R} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}$ is a known measurable mapping. We note that

- $\Phi(h_0) = W(h_0)$ when $\phi(h_0(x), x) = h_0(x)$;
- $\Phi(h_0) = V(h_0)$ when $\phi(h_0(x), x) = v_0(x)$ with $\partial_1 \phi(h_0(x), x) = 0$.

Assumption 2 (Functional). *The functional Φ satisfies*

- (a) *$\phi : \mathbb{R} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}$ is continuously differentiable with respect to its first argument.*
- (b) *Target Density: The target density f of X is absolutely continuous w.r.t. f_0 with uniformly bounded Radon-Nikodym derivative $\lambda := f/f_0$; consequently f is itself uniformly bounded on $\mathcal{X}$.*
- (c) *Regular Level Set: $h_0(\cdot) := \mu_0(\cdot, 1) - \mu_0(\cdot, 0)$ satisfies $\|\nabla_x h_0(x)\| \geq \underline{c} > 0$ on the level set $\{x \in \mathcal{X} : h_0(x) = 0\}$.*

2.2 Functional Derivatives via Generalized Leibniz rule

By the standard semiparametric theory on the estimation of functionals of nonparametric regression functions, the asymptotic property of the semiparametric plug-in estimator $\Phi(\hat{h})$ can be analyzed via the functional derivative of $\Phi(h)$ w.r.t. h_0 in the direction of $h - h_0$, i.e., writing $h_t := h_0 + t(h - h_0)$,

$$D_h \Phi(h_0)[h - h_0] := \left. \frac{d}{dt} \Phi(h_t) \right|_{t=0} = \left. \frac{d}{dt} \int \mathbb{1}\{h_t(x) \geq 0\} \phi(h_t(x), x) f(x) dx \right|_{t=0}, \quad (7)$$

$$D_h W(h_0)[h - h_0] := \left. \frac{d}{dt} W(h_t) \right|_{t=0} = \left. \frac{d}{dt} \int [h_t(x)]_+ f(x) dx \right|_{t=0} \quad (8)$$

$$D_h V(h_0)[h - h_0] := \left. \frac{d}{dt} V(h_t) \right|_{t=0} = \left. \frac{d}{dt} \int \mathbb{1}\{h_t(x) \geq 0\} v_0(x) f(x) dx \right|_{t=0}. \quad (9)$$

The presence of the ReLU/max function $[t]_+$ in $D_h W(h_0)$ and the indicator function $\mathbb{1}\{t \geq 0\}$ induces a point of nonsmoothness at $t = 0$, where $[t]_+$ is nondifferentiable and $\mathbb{1}\{t \geq 0\}$ is discontinuous. This complicates the calculation of the functional derivatives in (8) and (9), though to different degrees.

We now provide an overview of the key difference between the welfare and value functionals from the perspective of the generalized Leibniz rule, which has the following generic form regarding the total time derivative of integrals with changing integrand and changing region of integration:

$$\frac{d}{dt} \int_{\Omega_t} G_t(x) dx. \quad (10)$$

The generalized Leibniz rule, states that, under mild regularity conditions,

$$\frac{d}{dt} \int_{\Omega_t} G_t(x) dx = \underbrace{\int_{\Omega_t} \frac{\partial}{\partial t} G_t(x) dx}_{\text{(I)}} + \underbrace{\int_{\partial\Omega_t} \langle \mathbf{n}_t(x), \mathbf{v}_t(x) \rangle G_t(x) dS_{\partial\Omega_t}(x)}_{\text{(II)}} \quad (11)$$

where term (I) captures the effect of the change in the integrand $G_t(x)$ with the region of integration Ω_t held fixed, while term (II) captures the effect of the change in the region of integration Ω_t with the integrand $G_t(x)$ held fixed. The somewhat “nonstandard” term (II) warrants some more explanations: $\partial\Omega_t$ denotes the boundary of Ω_t , $\mathbf{n}_t(x)$ is the outward-pointing unit normal vector, $\mathbf{v}_t(x)$ is the velocity vector associated with the time movement in the $\partial\Omega_t$, and $S_{\partial\Omega_t}(x)$ denotes the surface measure on the boundary $\partial\Omega_t$. Note that, when x is one-dimensional and $\Omega_t = [a_t, b_t]$, (11) specializes to the standard Leibniz rule:

$$\frac{d}{dt} \int_{a_t}^{b_t} G_t(x) dx = \int_{a_t}^{b_t} \frac{\partial}{\partial t} G_t(x) dx + G_t(b_t) \frac{d}{dt} b_t - G_t(a_t) \frac{d}{dt} a_t.$$

We observe that $D_h \Phi(h_0)$, $D_h W(h_0)$ and $D_h V(h_0)$ are all of the form (10), with the

same parametrized region of integration

$$\Omega_t := \{x \in \mathbb{R}^{d_x} : h_t(x) \geq 0\} \quad , \quad \Omega_0 := \{x \in \mathbb{R}^{d_x} : h_0(x) \geq 0\}$$

and $\partial\Omega_0 = \{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}$ under mild regularity conditions on h_0 .

Applying the generalized Leibniz rule (11) to the functional $\Phi(h)$, we obtain:

$$\begin{aligned} D_h \Phi(h_0)[h - h_0] &= \underbrace{\int_{\Omega_0} \partial_1 \phi(h_0(x), x) (h(x) - h_0(x)) f(x) dx}_{\text{(I)}} \\ &\quad + \underbrace{\int_{\partial\Omega_0} \langle \mathbf{n}_0(x), \mathbf{v}_0(x) \rangle \phi(h_0(x), x) f(x) dS_{\partial\Omega_0}(x)}_{\text{(II)}} \end{aligned} \quad (12)$$

where the first term (I) is a full-dimensional Lebesgue integral (in $\mathbb{R}^{d_x}$), while the second term (II) is a lower-dimensional boundary integral. In particular, when $\phi(h_0(x), x) f(x)$ does not vanish on $\partial\Omega_0 = \{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}$, the second term (II) cannot be ignored, even though $\partial\Omega_0$ has Lebesgue measure zero in $\mathbb{R}^{d_x}$. In fact, under Assumption 2(c) (see, e.g., [Chen and Gao \(2026\)](#)), the second term (II) of (12) can be expressed as

$$\int_{\partial\Omega_0} \langle \mathbf{n}_0(x), \mathbf{v}_0(x) \rangle \phi(h_0(x), x) f(x) dS_{\partial\Omega_0}(x) = \int_{\partial\Omega_0} \frac{h(x) - h_0(x)}{\|\nabla_x h_0(x)\|} \phi(h_0(x), x) f(x) d\mathcal{H}^{d_x-1}(x), \quad (13)$$

where $\mathcal{H}^{d_x-1}$ denotes the $(d_x - 1)$ -dimensional Hausdorff measure (see [Chen and Gao \(2026\)](#)), which coincides with the $(d_x - 1)$ -dimensional Lebesgue measure in $\mathbb{R}^{d_x-1}$. Thus the boundary integral term (II) of (12) is a lower-dimensional integral functional that only extracts information about h_0 on a Lebesgue measure-0 set (in $\mathbb{R}^{d_x}$).

For the welfare functional $\Phi(h_0) = W(h_0)$, plugging $\phi(h_0(x), x) = h_0(x)$ into (12) yields

$$D_h W(h_0)[h - h_0] = \int \mathbb{1}\{h_0(x) \geq 0\} [h(x) - h_0(x)] f(x) dx, \quad (14)$$

where the term (II) vanishes since $\phi(h_0(x), x) = h_0(x) = 0$ on the boundary $\partial\Omega_0$, and the term (I) is a full-dimensional Lebesgue integral functional of $h - h_0$.

For the value functional $\Phi(h_0) = V(h_0)$, plugging $\phi(h_0(x), x) = v_0(x)$ with $\partial_1 \phi(h_0(x), x) = 0$ into (12) and (13) yields

$$D_h V(h_0)[h - h_0] = \int_{\{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}} \frac{(h(x) - h_0(x))}{\|\nabla_x h_0(x)\|} v_0(x) f(x) d\mathcal{H}^{d_x-1}(x), \quad (15)$$

which is nonzero as long as $v_0(x)f(x)$ does not vanish on the boundary $\partial\Omega_0 = \{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}$. For example, setting $v_0(x) \equiv 1$ yields $V(h_0) = \mathbb{P}_f(h_0(X_i) \geq 0)$, the share of population with nonnegative CATE, and the boundary integral term (II) does not vanish. In fact, as long as $v_0(x)f(x) \neq 0$ on the boundary $\partial\Omega_0 = \{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}$,

$D_h V(h_0)$ is a nonzero $(d_x - 1)$ -dimensional integral functional that only extracts information about h_0 on a Lebesgue measure-0 set (in $\mathbb{R}^{d_x}$), akin to a point evaluation of a nonparametric estimation.

2.3 Necessary Condition for $\sqrt{n}$ -Estimability

Let $L^2(f)$ denote the Hilbert space of square integrable (against f) functions with the inner product $\langle g, h \rangle_{2,f} := \int g(x) h(x) f(x) dx$. For the CATE function $h_0 \in L^2(f)$, it is well-known that a linear functional $L[h - h_0]$ is bounded (or equivalently, continuous) if and only if

$$\sup_{\nu \neq 0, \nu \in L^2(f)} \frac{|L[\nu(\cdot)]|^2}{E_f[|\nu(X)|^2]} < \infty$$

which is a necessary and sufficient condition for the existence of a Riesz representer $\nu^* \in L^2(f)$ such that

$$L[\nu] = \langle \nu^*, \nu \rangle_{2,f} \quad \text{for all } \nu \in L^2(f).$$

This in turn is a necessary condition for any plug-in estimator of the linear functional $L[\hat{h} - h_0] = \langle \nu^*, \hat{h} - h_0 \rangle_{2,f}$ to converge to zero at a $\sqrt{n}$ rate.

For the linear functional $L[\nu] = D_h W(h_0)[\nu]$ given in (14), we immediately see that $\nu_h^*(x) := \mathbb{1}\{h_0(x) \geq 0\}$ is the Riesz representer of the linear functional $D_h W(h_0)[\nu]$ in the Hilbert space $L^2(f)$, and that this Riesz representer has bounded norm

$$\|\nu_h^*\|^2 := \int \mathbb{1}^2\{h_0(x) \geq 0\} f(x) dx \leq 1,$$

and thus, by well-known results in, say, [Chen et al. \(2014\)](#) and [Chen and Pouzo \(2015\)](#), the linear functional $D_h W(h_0)[\nu]$ is a regular (i.e., $\sqrt{n}$ -estimable) functional under appropriate conditions.

In contrast, the linear functional $D_h V(h_0)[\nu]$ given in (15) is not a bounded linear functional on $L^2(f)$, and hence does not have a well-defined Riesz representer in the Hilbert space $L^2(f)$. Consequently, by Lemma 3.3 of [Chen and Pouzo \(2015\)](#), the functional V is an irregular functional that cannot be estimated at the $\sqrt{n}$ rate — provided the value functional is *nondegenerate*, i.e., the boundary weight $v_0 f$ does not vanish on the level set $\{x : h_0(x) = 0\}$ and h_0 crosses zero regularly there (Assumption 2(c)). This nondegeneracy is precisely what makes the boundary term (15) nonzero, and is formalized by the positive-boundary-integral condition of the minimax lower bound (Theorem 5). In degenerate cases — for instance if $v_0 f$ vanishes on the boundary or if h_0 is bounded away from zero — $V(h_0)$ may again be $\sqrt{n}$ -estimable or even trivial.

The above provides an intuitive explanation of why the welfare functional $W(h_0)$ is very special relative to general types of value functionals $V(h_0)$ or $\Phi(h_0)$, and clarifies why $W(h_0)$

could be $\sqrt{n}$ -estimable while others generally cannot. In subsequent sections, we provide formal conditions and theorems that establish the $\sqrt{n}$ -normality of plug-in estimators of the welfare functional $W(h_0)$, as well as the slower-than- $\sqrt{n}$ asymptotic normality for the value functional $V(h_0)$.

3 Estimation and Inference of the Welfare Functional

In this section, we focus on the welfare functional defined in (3), which can also be equivalently written as a functional of μ_0 as follows:

$$\begin{aligned} W(h_0) &:= \int [h_0(x)]_+ f(x) dx \\ &\equiv \overline{W}(\mu_0) := \int [\mu_0(x, 1) - \mu_0(x, 0)]_+ f(x) dx. \end{aligned} \tag{16}$$

We write out the two equivalent definitions of the functionals based on h_0 and μ_0 , since each representation has its own merit. The representation $W(h_0)$ based on the CATE function h_0 is clearer in terms of its interpretation: the condition $h_0(x) \geq 0$ is a direct optimal treatment assignment, and this representation has been adopted in previous work such as [Kitagawa and Tetenov \(2018\)](#). On the other hand, the representation $\overline{W}(\mu_0)$ based on μ_0 is clearer in terms of the underlying nonparametric regression function, which is notationally easier to work with in our subsequent semiparametric asymptotic analysis.

We consider two different setups for the estimation of $W(h_0)$, depending on how we treat the marginal density $f(x)$.

In the first setup, we treat f as known and the functional $W(\cdot)$ as a known transformation of h_0 . This is relevant in cases where the covariate density $f(x)$ of the target population is either configured or known/estimated from other sources than the training sample used to estimate CATE. For example, CATE may be estimated from a smaller pilot program (with sample size n), while the policymakers are contemplating implementing the policy on a statewide or nationwide basis with a much larger target population, whose covariate density may either be known at the population level or estimated from an alternative data source with much larger sample size $N \gg n$ (so that the sampling uncertainty in the estimation of f becomes negligible relative to that in the estimation of h_0).

In the second setup, we take f to be unknown and set it to f_0 , the covariate density in the underlying population of the training sample. In this case, f does not need to be explicitly estimated, but the integral in $W(\cdot)$ with respect to f can be naturally approximated via sample average in the training sample. This corresponds more closely to the “empirical welfare” as considered in [Kitagawa and Tetenov \(2018\)](#). We provide results for this setting as well.

3.1 Welfare Functional Under Known Density f

We start with the first setup, where the covariate density f is taken to be known and $W(\cdot)$ is treated as a deterministic known integral of h_0 w.r.t. f . In this case, we can define a simple nonparametric plug-in estimator of $W(h_0)$ as

$$W(\hat{h}) \equiv \overline{W}(\hat{\mu}) = \int [\hat{h}(x)]_+ f(x) dx.$$

We impose the following sufficient conditions.

Assumption 3. *Suppose that:*

- (a) *First-Stage Convergence:* $\|\hat{\mu} - \mu_0\|_\infty = o_p(n^{-1/4})$.
- (b) *Asymptotic linearity of the first stage:* the first-stage estimator $\hat{\mu}$ is asymptotically linear for μ_0 in the direction of the regular functional, in the sense that the bounded Riesz representer ν^* of $D_\mu \overline{W}(\mu_0)[\cdot]$ given in Theorem 1 satisfies

$$\sqrt{n} D_\mu \overline{W}(\mu_0)[\hat{\mu} - \mu_0] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \nu^*(X_i, D_i) \epsilon_i + o_p(1).$$

This is a property of the first-stage estimator alone (not of the welfare plug-in), and, because $\overline{W}$ is regular, it is satisfied by a broad range of nonparametric first stages — sieve/series least squares and other linear smoothers under undersmoothing, as well as Neyman-orthogonal (debiased) estimators. Lemma 1 below records concrete lower-level conditions that imply it.

- (c) *Stochastic equicontinuity (only for Theorem 2):* $\hat{\mu} - \mu_0$ belongs to a P -Donsker class with probability approaching one (for a linear sieve first stage this is implied by the conditions of Lemma 1 with $K_n \log K_n/n \rightarrow 0$).

Assumption 3 separates the asymptotic behavior of the plug-in $\overline{W}(\hat{\mu})$ into two distinct pieces and thereby avoids circularity: condition (a) controls the second-order *curvature* remainder of the nonlinear map $\mu \mapsto \overline{W}(\mu)$, while condition (b) is a property of the *first stage* $\hat{\mu}$, namely its asymptotic linearity for μ_0 . Neither presupposes the asymptotic normality of $\overline{W}(\hat{\mu})$, which is their consequence. Stating (b) at this higher level keeps Theorem 1 agnostic about the choice of first stage; the boundedness of ν^* established in Theorem 1 is precisely what makes $\overline{W}$ a regular ($\sqrt{n}$ -estimable) functional, so that (b) holds for many standard estimators (Lemma 1).

Theorem 1 ($\sqrt{n}$ -asymptotic normality for the welfare functional). *Under Assumptions 1 and 2, we have:*

$$\nu^*(x, d) := \mathbb{1}\{h_0(x) \geq 0\} \lambda(x) \left(\frac{d}{p_0(x)} - \frac{1-d}{1-p_0(x)} \right)$$

is the Riesz representer for the linear functional $D_\mu \overline{W}(\mu_0)[\cdot]$.

Further, if Assumption 3(a)(b) holds, we have:

$$\sqrt{n} \left(\overline{W}(\hat{\mu}) - \overline{W}(\mu_0) \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \nu^*(X_i, D_i) \epsilon_i + o_p(1).$$

Denote $\sigma_\epsilon^2(x, d) := \mathbb{E}[\epsilon_i^2 | X_i = x, D_i = d]$. Then:

$$\sqrt{n} \left(W(\hat{h}) - W(h_0) \right) \equiv \sqrt{n} \left(\overline{W}(\hat{\mu}) - \overline{W}(\mu_0) \right) \xrightarrow{d} \mathcal{N}(0, \sigma_W^2),$$

with

$$\sigma_W^2 := \mathbb{E} \left[(\nu^*(X_i, D_i) \epsilon_i)^2 \right] = \mathbb{E} \left[\mathbb{1} \{h_0(X_i) \geq 0\} \lambda^2(X_i) \left(\frac{\sigma_\epsilon^2(X_i, 1)}{p_0(X_i)} + \frac{\sigma_\epsilon^2(X_i, 0)}{1 - p_0(X_i)} \right) \right].$$

Lemma 1 (Lower-level sufficient conditions for Assumption 3(b)). *Let $\hat{\mu}$ be a linear (B-spline) sieve least squares estimator of μ_0 — for instance, the stacked sieve regression of Section 3.3 — satisfying the standard linear-sieve conditions of Chen and Gao (2026) (minimum Gram-matrix eigenvalue bounded away from zero, $\sup_x \|\psi^{(K)}(x)\| = O(\sqrt{K})$, $K \log K/n = o(1)$, and an L^∞ -bounded sieve projection operator), with the sieve dimension undersmoothed so that $\sqrt{n} K_n^{-s/d_x} \rightarrow 0$. Then Assumption 3(b) holds.*

Proof. Since the Riesz representer ν^* of $D_\mu \overline{W}(\mu_0)[\cdot]$ is bounded (Theorem 1), the sieve least squares asymptotic linear representation of Chen, Liao and Sun (2014); Chen and Pouzo (2015) gives

$$\sqrt{n} D_\mu \overline{W}(\mu_0)[\hat{\mu} - \mu_0] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \nu^*(X_i, D_i) \epsilon_i + o_p(1),$$

where the undersmoothing $\sqrt{n} K_n^{-s/d_x} \rightarrow 0$ ensures that the sieve approximation bias $\sqrt{n} D_\mu \overline{W}(\mu_0)[\Pi_{K_n} \mu_0 - \mu_0] = o(1)$ (using $\|\nu^*\| < \infty$ and $\|\Pi_{K_n} \mu_0 - \mu_0\|_2 = O(K_n^{-s/d_x})$, with Π_{K_n} the L^2 sieve projection), and the sieve Riesz representer converges to ν^* in L^2 . $\square$

More generally, beyond linear smoothers, condition (b) is delivered by Neyman-orthogonal/debiased first stages: the orthogonal-moment construction of Section 5.5 (Corollary 1) attains the same influence-function representation under the product-rate conditions there, dispensing with the Donsker requirement (c) altogether.

Theorem 1 suggests the following natural estimator for the asymptotic variance σ_W^2 :

$$\hat{\sigma}_W^2 := \frac{1}{n} \sum_{i=1}^n \mathbb{1} \{ \hat{h}(X_i) \geq 0 \} \lambda^2(X_i) \left(\frac{D_i}{\hat{p}^2(X_i)} + \frac{1 - D_i}{(1 - \hat{p}(X_i))^2} \right) \hat{\epsilon}_i^2$$

where $\hat{\epsilon}_i := Y_i - \hat{\mu}(X_i, D_i)$ and $\hat{p}$ is a first-stage estimator of the propensity score p_0 . We note that, in the special case where the conditional variance of the regression error does not depend on the treatment status, i.e., $\sigma_\epsilon^2(x, d) = \sigma_\epsilon^2(x)$, the asymptotic variance simplifies to

$\sigma_W^2 = \mathbb{E}[\mathbb{1}\{h_0(X_i) \geq 0\} \lambda^2(X_i) \sigma_\epsilon^2(X_i) / (p_0(X_i)(1 - p_0(X_i)))]$, which may equivalently be estimated by replacing the term in parentheses in $\hat{\sigma}_W^2$ with $1/(\hat{p}(X_i)(1 - \hat{p}(X_i)))$.

Remark 1. The variance estimator $\hat{\sigma}_W^2$ above involves the density ratio $\lambda = f/f_0$, which is directly computable only when the training-sample covariate density f_0 is known (in addition to the target density f). When f_0 is unknown, λ may be replaced by $f/\hat{f}_0$ with a nonparametric estimator $\hat{f}_0$ of f_0 ; alternatively, one may use the sieve Riesz representer variance estimator developed in Section 3.3 below, which sidesteps the explicit estimation of both λ and the propensity score p_0 .

3.2 Welfare Functional Under Unknown Density $f = f_0$

In certain cases, such as in Kitagawa and Tetenov (2018), one might be interested in the welfare functional under the original distribution of covariates F_0 , which may not be known or controlled. Given the random sample of $(Y_i, D_i, X_i)_{i=1}^n$ used to estimate the CATE h_0 , it is natural to use the sample mean $\frac{1}{n} \sum_{i=1}^n [\cdot]$ as an estimator of the population expectation $\int [\cdot] dF_0$, in which case a natural plug-in estimator of $W(h_0) \equiv \overline{W}(\mu_0)$ is given by

$$\hat{W}(\hat{h}) \equiv \hat{\overline{W}}(\hat{\mu}) := \frac{1}{n} \sum_{i=1}^n [\hat{\mu}(X_i, 1) - \hat{\mu}(X_i, 0)]_+ \equiv \frac{1}{n} \sum_{i=1}^n [\hat{h}(X_i)]_+.$$

Clearly, the approximation of the integral introduces an additional source of randomness, but the result established in the last subsection continues to be useful in this case.

Theorem 2. *Under Assumptions 1, 2, and 3,*

$$\begin{aligned} \sqrt{n}(\hat{W}(\hat{h}) - W(h_0)) &\equiv \sqrt{n}(\hat{\overline{W}}(\hat{\mu}) - \overline{W}(\mu_0)) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n ([h_0(X_i)]_+ - W(h_0) + \nu^*(X_i, D_i) \epsilon_i) + o_p(1) \xrightarrow{d} \mathcal{N}(0, \bar{\sigma}_W^2) \end{aligned}$$

where $\bar{\sigma}_W^2 := \mathbb{E} \left[\left([h_0(X_i)]_+ - W(h_0) + \nu^*(X_i, D_i) \epsilon_i \right)^2 \right] = \text{Var}([h_0(X_i)]_+) + \sigma_W^2$.

3.3 Inference Based on the Sieve Riesz Representer

The asymptotic variance formulas in Theorems 1 and 2 lead to analytic plug-in variance estimators that involve the density ratio $\lambda = f/f_0$ and the propensity score p_0 . As noted in Remark 1, the density ratio is directly computable only when the training-sample density f_0 is known. In this subsection we develop an alternative variance estimator based on the sieve Riesz representer, following Chen and Christensen (2018), which sidesteps the explicit estimation of both λ and p_0 . The same sieve construction also serves as the basis for the standard error of the value functional in Section 4.1.

For this purpose we take the first-stage estimator $\hat{\mu}$ to be a linear sieve (B-spline) least squares estimator. In practice, $\hat{\mu}(x, 1)$ is estimated with K_1 linear series in the treated subsample, while $\hat{\mu}(x, 0)$ is estimated separately with potentially different K_0 linear series in the untreated subsample. Since we treat D_i as a random variable, however, we cannot regard $\hat{\mu}(x, 1)$ and $\hat{\mu}(x, 0)$ as two completely separate nonparametric estimators with exogenously given sample sizes. Instead, we treat $\hat{\mu}$ as the least squares estimator of the stacked regression

$$Y_i = D_i \psi^{(K_1)}(X_i)' \beta_1 + (1 - D_i) \psi^{(K_0)}(X_i)' \beta_0 + u_i \quad (17)$$

where $\psi^{(K_1)}(x) = (\psi_1(x), \dots, \psi_{K_1}(x))'$ and $\psi^{(K_0)}(x) = (\psi_{K_1+1}(x), \dots, \psi_{K_1+K_0}(x))'$ denote the B-spline basis functions used to estimate $\mu_0(x, 1)$ and $\mu_0(x, 0)$, respectively, with sieve dimensions K_1 and K_0 . Define the stacked regressor vector $\bar{\psi}(x, d)$ such that its k th component equals $d\psi_k(x)$ for $k \in \{1, \dots, K_1\}$ and $(1-d)\psi_k(x)$ for $k \in \{K_1+1, \dots, K_1+K_0\}$, and let $K_n := K_1 + K_0$.

Following [Chen and Christensen \(2018\)](#), specifically equations (6) and (7), the sieve variance estimator for the welfare functional takes the sandwich form

$$\hat{\sigma}_{W,sv}^2 := D_\mu \bar{W}(\hat{\mu}) [\bar{\psi}]' \hat{\Omega} D_\mu \bar{W}(\hat{\mu}) [\bar{\psi}],$$

where $\hat{\Omega}$ is the heteroskedasticity-robust estimator of the asymptotic covariance matrix of $\sqrt{n}(\hat{\beta} - \beta_0)$ for the stacked OLS coefficients $\hat{\beta} := (\hat{\beta}_1', \hat{\beta}_0')'$ in (17), namely the sandwich

$$\hat{\Omega} := \hat{G}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 \bar{\psi}_i \bar{\psi}_i' \right) \hat{G}^{-1}, \quad \hat{G} := \frac{1}{n} \sum_{i=1}^n \bar{\psi}_i \bar{\psi}_i', \quad \bar{\psi}_i := \bar{\psi}(X_i, D_i), \quad \hat{u}_i := Y_i - \hat{\mu}(X_i, D_i),$$

following [Chen and Christensen \(2018\)](#) and the sieve variance estimator of [Chen and Gao \(2026\)](#); this scaling is consistent with the $1/\sqrt{n}$ -normalized score bootstrap of [Appendix D.6](#). Because $\hat{\Omega}$ is heteroskedasticity-robust, it automatically accommodates treatment-arm-specific conditional error variances. By (14), the pathwise derivative evaluated at the sieve basis is the K_n -vector

$$D_\mu \bar{W}(\hat{\mu}) [\bar{\psi}] = \begin{pmatrix} \int_{\{x \in \mathcal{X} : \hat{h}(x) \geq 0\}} \psi^{(K_1)}(x) f(x) dx \\ - \int_{\{x \in \mathcal{X} : \hat{h}(x) \geq 0\}} \psi^{(K_0)}(x) f(x) dx \end{pmatrix},$$

where $\hat{h}(x) := \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$.

Crucially, the pathwise derivative $D_\mu \bar{W}(\hat{\mu}) [\bar{\psi}]$ is a *full-dimensional* Lebesgue integral over the upper-contour set $\{\hat{h} \geq 0\}$, rather than a lower-dimensional Hausdorff integral over the boundary $\{\hat{h} = 0\}$. Consequently it can be computed directly by numerical integration over Sobol points drawn from F , and *no numerical differentiation* (i.e., no ϵ -band approximation of the level set) is required. This reflects the regularity of the welfare functional established in [Section 2](#), and contrasts with the value functional of [Section 4.1](#) below, whose

standard error involves a boundary integral that does require numerical differentiation.

Under the conditions of Theorem 1 together with the consistency of $\hat{\Omega}$ as in Chen and Christensen (2018), we have $\hat{\sigma}_{W,\text{sv}}^2/\sigma_W^2 \xrightarrow{p} 1$, so that the studentized statistic is asymptotically standard normal,

$$\frac{\sqrt{n} \left(\overline{W}(\hat{\mu}) - \overline{W}(\mu_0) \right)}{\hat{\sigma}_{W,\text{sv}}} \xrightarrow{d} \mathcal{N}(0, 1),$$

and the confidence interval $\left[\overline{W}(\hat{\mu}) \pm 1.96 \hat{\sigma}_{W,\text{sv}}/\sqrt{n} \right]$ has asymptotically correct coverage.

In the unknown-density case $f = f_0$ of Section 3.2, the integrals defining $D_\mu \overline{W}(\hat{\mu})[\overline{\psi}]$ are taken with respect to f_0 and approximated by the corresponding sample averages over the training data, e.g.

$$\int_{\{\hat{h} \geq 0\}} \psi^{(K_1)}(x) f_0(x) dx \approx \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\hat{h}(X_i) \geq 0\} \psi^{(K_1)}(X_i).$$

By Theorem 2, the asymptotic variance then acquires the additional term $\text{Var}([h_0(X_i)]_+)$, which is estimated by the sample variance of $[\hat{h}(X_i)]_+$, yielding

$$\hat{\sigma}_{W,\text{sv}}^2 := \widehat{\text{Var}}\left([\hat{h}(X_i)]_+\right) + \hat{\sigma}_{W,\text{sv}}^2.$$

Simulation results based on this sieve variance estimator are reported in Appendix C.5.

Beyond the plug-in approach, the welfare functional also admits a debiased machine learning estimator that accommodates generic machine-learning first-stage estimators of the CATE. Because the underlying construction — based on the sieve Riesz representer — applies uniformly whether the functional is regular or irregular, we develop it in the standalone Section 5, where the welfare functional is treated in Section 5.5.

4 Estimation and Inference of the Value Functional

4.1 Value Functional Under Known Density f

We now turn to the general value functional given by

$$\begin{aligned} V(h_0) &:= \int \mathbb{1}\{h_0(x) \geq 0\} v_0(x) f(x) dx \\ &\equiv \overline{V}(\mu_0) := \int \mathbb{1}\{\mu_0(x, 1) - \mu_0(x, 0) \geq 0\} v_0(x) f(x) dx. \end{aligned}$$

The simple plug-in estimators are defined as

$$\begin{aligned} V(\hat{h}) &:= \int \mathbb{1}\{\hat{h}(x) \geq 0\} v_0(x) f(x) dx \\ &\equiv \overline{V}(\hat{\mu}) := \int \mathbb{1}\{\hat{\mu}(x, 1) - \hat{\mu}(x, 0) \geq 0\} v_0(x) f(x) dx. \end{aligned}$$

Under Assumptions 1 and 2, the functional derivative of $\bar{V}(\mu_0)$ is given by

$$D_\mu \bar{V}(\mu_0)[\nu] := \int_{\{x \in \mathbb{R}^{d_x} : h_0(x)=0\}} \frac{(\nu(x, 1) - \nu(x, 0))}{\|\nabla_x h_0(x)\|} v_0(x) f(x) d\mathcal{H}^{d_x-1}(x).$$

As explained in Section 2, $V(h_0) \equiv \bar{V}(\mu_0)$ is generally not $\sqrt{n}$ -estimable: in particular, $D_\mu \bar{V}(\mu_0)[\cdot]$ is not a bounded (or continuous) linear functional on $L^2(f)$, which means that it does not have a Riesz representer on the whole Hilbert space $L^2(f)$. Nevertheless, the sieve Riesz representer is well-defined (see Chen and Gao (2026)), which will be the key ingredient of the sieve variance estimator for $\bar{V}(\hat{\mu}) - \bar{V}(\mu_0)$. In particular we need to study the asymptotic property of the plug-in estimator of the submanifold integral of form (15), which has been studied in Chen and Gao (2026) using linear series (B-spline) first stage: in particular, Section 4.3 of Chen and Gao (2026) analyzes the integral on upper contour set of the form $V(h_0)$ specifically. We use the result in Chen and Gao (2026) without repeating it here, but focus on the adaptation required for the standard error computation.

Assumption 4. *Suppose that:*

- (a) *Weight and curvature regularity: $\|\nabla_x^2 h_0(x)\| \leq M < \infty$ on $\mathcal{X}$, and the surface weight $v_0 f$ together with its gradient $\nabla_x(v_0 f)$ is uniformly bounded on $\mathcal{X}$ (the analog, for the weight $w = v_0 f$, of the regularity conditions for the upper-contour integral in Chen and Gao, 2026). Together with the regular level set in Assumption 2(c), this matches their Assumption on $V(h_0)$.*
- (b) *Linearization remainder: $\|\hat{\mu} - \mu_0\|_\infty \|\nabla(\hat{\mu} - \mu_0)\|_\infty = o_p\left(\sqrt{\frac{1}{n} K_n^{\frac{1}{d_x}}}\right)$.*
- (c) *Error moments: $0 < \inf_{x,d} \sigma_\epsilon^2(x, d) \leq \sup_{x,d} \sigma_\epsilon^2(x, d) < \infty$; the Lindeberg condition $\sup_{x,d} \mathbb{E}[\epsilon_i^2 \mathbb{1}\{|\epsilon_i| > c\} | X_i = x, D_i = d] \rightarrow 0$ as $c \rightarrow \infty$; and, for consistency of the variance estimator, $\mathbb{E}[|\epsilon_i|^{2+\delta}] < \infty$ for some $\delta > 2d_x/(2s - (d_x - 1))$ (well-defined since the maintained smoothness $s > d_x$ implies $2s - (d_x - 1) > 0$).*
- (d) *Linear sieves and undersmoothing: for $d \in \{0, 1\}$, the B-spline sieve $\psi^{(K_d)}$ satisfies the standard linear-sieve conditions of Chen and Gao (2026) — the Gram matrix has minimum eigenvalue bounded away from zero, $\sup_{x \in \mathcal{X}} \|\psi^{(K_d)}(x)\| = O(\sqrt{K_d})$, $K_d \log K_d/n = o(1)$, the sup-norm sieve projection operator is $O_p(1)$, and the sieve approximation error for $\mu_0(\cdot, d)$ is $O(K_d^{-s/d_x})$ — and the sieve dimension is undersmoothed, $K_n \asymp n^{\frac{d_x}{2s+1}} [\log n]^\kappa$ for some $\kappa \in (0, 1)$.*

Theorem 3. *Let $\hat{\mu}$ be the stacked linear sieve least squares estimator (17) of μ_0 , and suppose that Assumptions 1, 2 and 4 hold with Hölder smoothness $s > d_x$. Then, specializing the*

upper-contour integral results of [Chen and Gao \(2026\)](#) (their Lemma 3, Theorem 5 and Propositions 1–2) to the CATE function $h_0 = \mu_0(\cdot, 1) - \mu_0(\cdot, 0)$:

$$\frac{\sqrt{n} \left(\bar{V}(\hat{\mu}) - \bar{V}(\mu_0) \right)}{\sigma_{V,n}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \text{with } \sigma_{V,n}^2 \asymp K_n^{\frac{1}{d_x}}.$$

Remark 2 (Convergence Rate and Choice of K_n). Theorem 3 implies that the plug-in estimator converges at the rate $\sigma_{V,n}/\sqrt{n} \asymp \sqrt{K_n^{1/d_x}/n}$. With the rate-optimal sieve dimension $K_n \asymp n^{\frac{d_x}{2s+1}}$ (up to a logarithmic factor), this rate becomes $r_n^* := n^{-\frac{s}{2s+1}}$, the optimal rate for the estimation of a 1-dimensional nonparametric regression with smoothness s . Theorem 5 in Section 4.3 below shows that r_n^* is in fact the fastest possible rate among all estimators of $V(h_0)$, so that the plug-in estimator is minimax rate-optimal.

We now discuss the adaptation required to compute an estimator of $\sigma_{V,n}^2$. We employ the same stacked sieve least squares first stage (17), stacked regressor vector $\bar{\psi}(x, d)$, and estimated OLS covariance matrix $\hat{\Omega}$ introduced for the welfare functional in Section 3.3; in particular, $\hat{\mu}(x, 1)$ and $\hat{\mu}(x, 0)$ are estimated jointly through the stacked regression (17) rather than as two separate nonparametric fits, since D_i is treated as random. Following [Chen and Christensen \(2018\)](#), specifically equations (6) and (7), the sieve variance estimator for the value functional takes the analogous sandwich form

$$\hat{\sigma}_V^2 := D_\mu \bar{V}(\hat{\mu}) [\bar{\psi}]' \hat{\Omega} D_\mu \bar{V}(\hat{\mu}) [\bar{\psi}],$$

where $\hat{\Omega}$ is as defined in Section 3.3.

In contrast to the full-dimensional integral for the welfare functional in Section 3.3, the pathwise derivative $D_\mu \bar{V}(\hat{\mu}) [\bar{\psi}]$ for the value functional is a lower-dimensional boundary (Hausdorff) integral, given by

$$D_\mu \bar{V}(\hat{\mu}) [\bar{\psi}] = \begin{pmatrix} \int_{\{x \in \mathcal{X} : \hat{h}(x)=0\}} \frac{\psi^{(K_1)}(x)}{\|\nabla_x \hat{h}(x)\|} v_0(x) f(x) d\mathcal{H}^{d_x-1}(x) \\ - \int_{\{x \in \mathcal{X} : \hat{h}(x)=0\}} \frac{\psi^{(K_0)}(x)}{\|\nabla_x \hat{h}(x)\|} v_0(x) f(x) d\mathcal{H}^{d_x-1}(x) \end{pmatrix},$$

which can be approximated via

$$\hat{D}_\mu \bar{V}(\hat{\mu}) [\bar{\psi}] = \begin{pmatrix} \frac{1}{2\epsilon} \int_{\{x \in \mathcal{X} : -\epsilon < \hat{h}(x) < \epsilon\}} \psi^{(K_1)}(x) v_0(x) f(x) dx \\ - \frac{1}{2\epsilon} \int_{\{x \in \mathcal{X} : -\epsilon < \hat{h}(x) < \epsilon\}} \psi^{(K_0)}(x) v_0(x) f(x) dx \end{pmatrix},$$

based on the mathematical result² that

$$\lim_{\epsilon \searrow 0} \frac{1}{2\epsilon} \int_{\{x \in \mathcal{X} : -\epsilon < h(x) < \epsilon\}} \omega(x) dx = \int_{\{x \in \mathcal{X} : h(x)=0\}} \frac{\omega(x)}{\|\nabla_x h(x)\|} d\mathcal{H}^{d_x-1}(x).$$

²See Theorem 3.13.(iii) of [Evans and Gariepy \(2015\)](#).

4.2 Value Functional Under Unknown Density $f = f_0$

We now consider the case where $F = F_0$ and the population expectation $\mathbb{E}[\cdot]$ is estimated by the sample average $\frac{1}{n} \sum_{i=1}^n [\cdot]$, and seek to characterize the asymptotic behavior of the natural plug-in estimator of $V(h_0) \equiv \bar{V}(\mu_0)$ given by

$$\begin{aligned}\hat{V}(\hat{h}) &:= \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{\hat{h}(X_i) \geq 0\}} v_0(X_i) \\ &\equiv \hat{\bar{V}}(\hat{\mu}) := \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{\hat{\mu}(X_i, 1) - \hat{\mu}(X_i, 0) \geq 0\}} v_0(X_i).\end{aligned}$$

It turns out that the additional error in the approximation of $V(\hat{h})$ by $\hat{V}(\hat{h})$ is asymptotically negligible relative to $V(\hat{h}) - V(h_0)$, which converges at a slower-than- $\sqrt{n}$ rate.

Theorem 4. *Suppose the conditions of Theorem 3 hold and that v_0 is uniformly bounded. Then the sample-average plug-in estimator $\hat{V}(\hat{h})$ is asymptotically normal at the same rate and with the same normalization as the integral plug-in $\bar{V}(\hat{\mu})$ of Theorem 3: with $\sigma_{V,n}^2 \asymp K_n^{1/d_x}$,*

$$\frac{\sqrt{n}(\hat{V}(\hat{h}) - V(h_0))}{\sigma_{V,n}} \xrightarrow{d} \mathcal{N}(0, 1),$$

the additional error from replacing the integral with respect to f_0 by a sample average being asymptotically negligible relative to the slower-than- $\sqrt{n}$ term $\bar{V}(\hat{\mu}) - V(h_0)$.

4.3 Minimax Rate Optimality

The slower-than- $\sqrt{n}$ rate in Theorem 3 raises a natural question: could a different estimator — say, one based on debiased machine learning or smoothing techniques — recover the value functional at a faster rate? Adapting the minimax lower bound theory in Chen and Gao (2026) to the present treatment assignment setting, we show that the answer is negative: the rate $r_n^* = n^{-\frac{s}{2s+1}}$ is the fastest possible convergence rate among *all* estimators of $V(h_0)$.

Theorem 5 (Minimax Lower Bound for the Value Functional). *Suppose that Assumptions 1 and 2 hold. Furthermore, suppose that: (i) the training-sample covariate density f_0 is uniformly bounded away from zero and infinity on $\mathcal{X} = [0, 1]^{d_x}$; (ii) $\inf_{x \in \mathcal{X}, d \in \{0, 1\}} \sigma_\epsilon^2(x, d) > 0$; and (iii) v_0 is uniformly bounded with $\int_{\{x \in \mathcal{X}: h_0(x)=0\}} (v_0(x) f(x))^2 d\mathcal{H}^{d_x-1}(x) > 0$. Then $r_n^* = n^{-\frac{s}{2s+1}}$ is a minimax lower bound rate for the estimation of $\theta_0(P) := V(h_0)$, i.e.,*

$$\liminf_{n \rightarrow \infty} \inf_{\tilde{\theta}} \sup_P \mathbb{E}_P \left[n^{\frac{2s}{2s+1}} (\tilde{\theta} - \theta_0(P))^2 \right] \geq c > 0,$$

where the supremum is taken over all joint distributions P of (Y_i, D_i, X_i) satisfying Model (5) and conditions (i)–(iii) with $\mu_0(\cdot, d) \in \Lambda_c^s(\mathcal{X})$ for $d \in \{0, 1\}$, and the infimum is taken

over all estimators $\tilde{\theta}$ based on the training data $\{(Y_i, D_i, X_i)\}_{i=1}^n$.

Theorem 5, together with Remark 2, establishes that the plug-in estimator $\bar{V}(\hat{\mu})$ is minimax rate-optimal. It also implies that the slower-than- $\sqrt{n}$ rate for the value functional is not an artifact of the plug-in approach: no estimation technique, including debiased machine learning or smoothing-based methods, can recover $V(h_0)$ at a rate faster than $n^{-\frac{s}{2s+1}}$, in sharp contrast to the $\sqrt{n}$ -estimability of the welfare functional $W(h_0)$ established in Section 3.

4.4 Quadratic Debiasing

The minimax rate optimality of the simple plug-in estimator $\bar{V}(\hat{\mu})$ comes with a caveat: it requires the relatively strong smoothness condition $s \geq d_x$ (see Theorem 5 in Chen and Gao, 2026), which becomes restrictive as the covariate dimension d_x grows. Chen and Gao (2026) introduce two debiased sieve estimators, a split-sample (SS) estimator and a leave-one-out (LOO) estimator, that attain the minimax-optimal rate r_n^* under the substantially weaker smoothness requirement $s > (d_x + 1)/2$. For instance, with $d_x = 2$ covariates as in our simulations and empirical application, the plug-in estimator requires $s > 2$ while the debiased estimators require only $s > 3/2$. Both debiased estimators work by subtracting an estimate of the “diagonal” part of the quadratic remainder in the expansion of $V(\hat{h})$ around h_0 . We now adapt them to the treatment assignment setting.

It is instructive to contrast this device with the sieve-Riesz debiasing developed later in Section 5 (specialized to the welfare functional in Section 5.5). For the regular welfare functional, the existence of a global influence function makes it amenable to the classical orthogonal-moment / DML construction, whose first-order correction both accommodates generic machine-learning first stages and preserves the $\sqrt{n}$ rate. For the irregular value functional, by contrast, no global influence function exists, so first-order orthogonalization is unavailable on the full Hilbert space; the SS and LOO estimators instead target the *second-order* (quadratic) remainder, which is what limits the rate, and thereby relax the smoothness requirement while retaining the (necessarily slower-than- $\sqrt{n}$) minimax-optimal rate.

For the SS estimator, we randomly split the training sample into two halves and run the stacked sieve regression (17) on each half, yielding first-stage estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ and the corresponding CATE estimators $\hat{h}_j(x) := \hat{\mu}_j(x, 1) - \hat{\mu}_j(x, 0)$ for $j = 1, 2$. Writing $\bar{h} := (\hat{h}_1 + \hat{h}_2)/2$, the SS estimator is defined as

$$\hat{\theta}_{SS} := V(\bar{h}) - \frac{1}{8}D^2V(\bar{h})[\hat{h}_1 - \hat{h}_2, \hat{h}_1 - \hat{h}_2], \quad (18)$$

where D^2V denotes the second-order pathwise derivative of V , characterized in closed form

in [Chen and Gao \(2026\)](#) using the calculus of moving submanifolds.

For the LOO estimator, let $\bar{\psi}_i := \bar{\psi}(X_i, D_i)$ denote the stacked regressor vector in (17), and let $a(x) := (\psi^{(K_1)}(x)', -\psi^{(K_0)}(x)')'$, so that $\hat{h}(x) = a(x)' \hat{\beta}$ with $\hat{\beta} := (\hat{\beta}_1', \hat{\beta}_0')'$. Let $\hat{G} := \frac{1}{n} \sum_{i=1}^n \bar{\psi}_i \bar{\psi}_i'$. The LOO estimator is defined as

$$\hat{\theta}_{LOO} := V(\hat{h}) - \frac{1}{2n^2} \sum_{i=1}^n D^2V(\hat{h}) [\hat{s}_i, \hat{s}_i] (\hat{\epsilon}_i^{(-i)})^2, \quad (19)$$

where $\hat{s}_i(x) := a(x)' \hat{G}^{-1} \bar{\psi}_i$ captures the influence of observation i on the CATE estimate $\hat{h}(x)$, and $\hat{\epsilon}_i^{(-i)} := Y_i - \hat{\mu}^{(-i)}(X_i, D_i)$ is the leave-one-out residual computed with the stacked sieve regression estimated on the sample excluding observation i . The constants $\frac{1}{8}$ in (18) and $\frac{1}{2n^2}$ in (19), the influence term $\hat{s}_i$, and the second functional derivative D^2V are exactly those of the upper-contour-integral specialization in [Chen and Gao \(2026\)](#) (their Theorem 5 and the accompanying split-sample and leave-one-out construction, with weight $w = v_0 f$ and submanifold dimension $m = d_x - 1$); the only adaptation to the treatment-assignment setting is the stacked-regression structure (17) of the CATE first stage, under which $\hat{h}(x) = a(x)' \hat{\beta}$.

Theorem 6 (Debiased Estimation of the Value Functional). *Suppose that the conditions of Theorem 3 hold with the smoothness requirement $s > d_x$ weakened to $s > (d_x + 1)/2$, and let the sieve dimension satisfy $K_n \asymp n^{\frac{d_x}{2s+1}}$ up to a logarithmic factor. Then, for $\tilde{\theta} \in \{\hat{\theta}_{SS}, \hat{\theta}_{LOO}\}$:*

$$\tilde{\theta} - \theta_0 = O_p(r_n^*), \quad \text{and} \quad \frac{\sqrt{n}(\tilde{\theta} - \theta_0)}{\sigma_{V,n}} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\sigma_{V,n}^2 \asymp K_n^{\frac{1}{d_x}}$ is the same sieve variance as in Theorem 3. Moreover, the sieve variance estimator $\hat{\sigma}_V^2$ of Section 4.1 remains consistent, so that the same studentized confidence intervals apply with $\tilde{\theta}$ in place of $\bar{V}(\hat{\mu})$.

Remark 3 (Numerical Implementation of D^2V). In keeping with the numerical integration/differentiation approach of Section 4.1, the quadratic form $D^2V(\hat{h})[v, v]$ may be computed via the second-order central difference

$$D^2V(\hat{h})[v, v] \approx \frac{V(\hat{h} + \delta v) - 2V(\hat{h}) + V(\hat{h} - \delta v)}{\delta^2}$$

for a small step size $\delta > 0$, with each evaluation of V computed by Sobol-point numerical integration as before. This avoids the explicit computation of the curvature terms in the closed-form expression of D^2V in [Chen and Gao \(2026\)](#).

A complementary debiasing device that instead accommodates *generic* machine-learning first stages, built on the sieve Riesz representer, applies uniformly to both the welfare and value functionals. We develop it in the standalone Section 5, where the value functional is

treated in Section 5.6.

5 Debiased Machine Learning via the Sieve Riesz Representer

Sections 3 and 4 analyzed the welfare and value functionals through plug-in estimators, the former $\sqrt{n}$ -estimable and the latter only at the slower minimax rate $n^{-s/(2s+1)}$. We now show that a *single* debiased machine learning (DML) procedure covers both, and accommodates generic machine-learning first-stage estimators of the CATE. The construction rests on the *sieve Riesz representer* of the functional's pathwise derivative: unlike a global influence function, the sieve Riesz representer is always well-defined on the finite-dimensional sieve space, whether or not the functional is regular. The limiting behavior of the resulting estimator is governed entirely by the norm $\sigma_n := \|v_{K_n}^*\|_{sd}$ of this representer, and the welfare and value functionals emerge as the two regimes in which σ_n stays bounded (regular, $\sqrt{n}$ rate) or diverges (irregular, slower rate). Throughout we work, as in Sections 3.1 and 4.1, under a known target density f ; the unknown-density case is handled exactly as in Section 3.2 and noted where it differs.

5.1 A Generic Functional and Its Augmented Moment

Let $\Theta(\mu)$ denote a functional of the regression μ whose value $\theta_0 := \Theta(\mu_0)$ is the object of interest; both the welfare functional $\bar{W}$ of Section 3 and the value functional $\bar{V}$ of Section 4 are special cases. Let $\mathcal{V}$ denote the Hilbert space of functions $v : \mathcal{X} \times \{0, 1\} \rightarrow \mathbb{R}$ with $\mathbb{E}[v^2(X_i, D_i)] < \infty$, equipped with the inner product $\langle v_1, v_2 \rangle := \mathbb{E}[v_1(X_i, D_i) v_2(X_i, D_i)]$, induced norm $\|\cdot\|$, and standard-deviation norm $\|v\|_{sd}^2 := \mathbb{E}[v^2(X_i, D_i) \sigma_\epsilon^2(X_i, D_i)]$, where $\sigma_\epsilon^2(x, d) := \mathbb{E}[\epsilon_i^2 | X_i = x, D_i = d]$.

Given any $v \in \mathcal{V}$, the locally robust (augmented) functional associated with Θ ,

$$\tilde{\Theta}(\mu, v) := \Theta(\mu) + \mathbb{E}[v(X_i, D_i)(Y_i - \mu(X_i, D_i))], \quad (20)$$

satisfies $\tilde{\Theta}(\mu_0, v) = \Theta(\mu_0) = \theta_0$ for every v , since $\mathbb{E}[\epsilon_i | X_i, D_i] = 0$. Its functional derivative with respect to μ in the direction ν is

$$D_\mu \tilde{\Theta}(\mu_0, v)[\nu] = D_\mu \Theta(\mu_0)[\nu] - \langle v, \nu \rangle. \quad (21)$$

If $D_\mu \Theta(\mu_0)[\cdot]$ admits a Riesz representer $v^* \in \mathcal{V}$ — as it does for the regular welfare functional — then setting $v = v^*$ annihilates (21) for all $\nu \in \mathcal{V}$, the Neyman-orthogonality property underlying classical DML (Chernozhukov, Chetverikov, Demirer, Duflo, Hansen,

Newey and Robins, 2018). For an irregular functional such as $\bar{V}$, no such v^* exists in $\mathcal{V}$ (Section 2.3); we instead annihilate the derivative *locally*, on the sieve space.

5.2 The Sieve Riesz Representer

Let $\bar{\psi}(x, d)$ be the stacked sieve regressor of Section 3.3, with $K_n = K_1 + K_0$; let $\mathcal{V}_{K_n} \subset \mathcal{V}$ denote the linear span of its components, and Π_{K_n} the $\langle \cdot, \cdot \rangle$ -orthogonal projection from $\mathcal{V}$ onto $\mathcal{V}_{K_n}$. On the finite-dimensional space $\mathcal{V}_{K_n}$, the Riesz representer of $D_\mu \Theta(\mu_0)[\cdot]$ is *always* well-defined and given in closed form by

$$v_{K_n}^*(x, d) := \bar{\psi}(x, d)' R_{K_n}^{-1} D_\mu \Theta(\mu_0) [\bar{\psi}], \quad R_{K_n} := \mathbb{E} [\bar{\psi}(X_i, D_i) \bar{\psi}(X_i, D_i)'], \quad (22)$$

which satisfies $D_\mu \Theta(\mu_0)[\nu] = \langle v_{K_n}^*, \nu \rangle$ for all $\nu \in \mathcal{V}_{K_n}$. Consequently $D_\mu \tilde{\Theta}(\mu_0, v_{K_n}^*)[\nu] \equiv 0$ for all $\nu \in \mathcal{V}_{K_n}$: the first-order bias is annihilated on the sieve space, with $v_{K_n}^*$ playing the role of a *local influence function*. Its sample analog replaces R_{K_n} and $D_\mu \Theta(\mu_0) [\bar{\psi}]$ by their empirical and numerical counterparts,

$$\hat{v}_{K_n}^*(x, d) := \bar{\psi}(x, d)' \hat{R}_{K_n}^{-1} \hat{D}_\mu \Theta(\hat{\mu}) [\bar{\psi}], \quad \hat{R}_{K_n} := \frac{1}{n} \sum_{i=1}^n \bar{\psi}(X_i, D_i) \bar{\psi}(X_i, D_i)', \quad (23)$$

where $\hat{D}_\mu \Theta(\hat{\mu}) [\bar{\psi}]$ is the (numerical) pathwise derivative already computed for the sieve variance estimators of Sections 3.3 and 4.1; constructing $\hat{v}_{K_n}^*$ thus requires no ingredients beyond those. The key point is that (22)–(23) are well-defined *regardless of whether Θ is regular*: regularity governs only the magnitude of $\sigma_n := \|v_{K_n}^*\|_{sd}$, not the existence of $v_{K_n}^*$.

5.3 The Cross-Fitted Sieve-Debiased Estimator

We combine the local influence function with cross-fitting to accommodate generic machine-learning first stages. Let $(I_\kappa)_{\kappa=1}^{\mathcal{K}}$ be a random partition of $\{1, \dots, n\}$ into $\mathcal{K} < \infty$ folds. For each κ , let $\hat{\mu}_{-\kappa}$ be a generic machine-learning estimator of μ_0 and $\hat{v}_{K_n, -\kappa}^*$ the sieve Riesz representer (23), both computed on the subsample excluding fold κ . The cross-fitted sieve-debiased estimator of $\theta_0 = \Theta(\mu_0)$ is

$$\tilde{\theta} := \frac{1}{\mathcal{K}} \sum_{\kappa=1}^{\mathcal{K}} \left(\Theta(\hat{\mu}_{-\kappa}) + \frac{\mathcal{K}}{n} \sum_{i \in I_\kappa} \hat{v}_{K_n, -\kappa}^*(X_i, D_i) (Y_i - \hat{\mu}_{-\kappa}(X_i, D_i)) \right). \quad (24)$$

The correction annihilates the first-order estimation error of $\hat{\mu}_{-\kappa}$ on the sieve space, which is what allows $\tilde{\theta}$ to tolerate generic first stages $\hat{\mu}_{-\kappa}$ — e.g. random forests, gradient boosting, or neural networks — that need *not* be the stacked sieve least squares estimator of (17), but must satisfy the high-level local-robustness conditions of Appendix B (Assumption 5). We caution that these conditions — in particular the sieve-approximation-bias and product-rate

requirements — may be nontrivial to verify for specific learners.

5.4 Asymptotic Normality

The following result establishes asymptotic normality of $\tilde{\theta}$ for a generic functional Θ , with the rate determined by $\sigma_n = \|v_{K_n}^*\|_{sd}$. Its proof, the precise local-robustness conditions, and the Hilbert-space construction are collected in Appendix B.

Theorem 7 (DML via the Sieve Riesz Representer). *Suppose Assumptions 1 and 2 hold, together with the local-robustness conditions of Appendix B (Assumption 5). Let $\sigma_n^2 := \|v_{K_n}^*\|_{sd}^2$. Then the cross-fitted sieve-debiased estimator (24) admits the asymptotically linear representation*

$$\frac{\sqrt{n}(\tilde{\theta} - \theta_0)}{\sigma_n} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i, D_i)}{\sigma_n} \epsilon_i + o_p(1) \xrightarrow{d} \mathcal{N}(0, 1).$$

Moreover, if $\hat{\sigma}_n^2$ is ratio-consistent for σ_n^2 , i.e. $\hat{\sigma}_n^2/\sigma_n^2 \xrightarrow{p} 1$ (as holds, e.g., for the sieve sandwich variance of Section 3.3), then $\sqrt{n}(\tilde{\theta} - \theta_0)/\hat{\sigma}_n \xrightarrow{d} \mathcal{N}(0, 1)$. Ratio-consistency is the relevant notion because σ_n^2 may diverge in the irregular case.

Whether $\tilde{\theta}$ is $\sqrt{n}$ -consistent is determined entirely by the sieve Riesz norm σ_n . When Θ is regular, $D_\mu \Theta(\mu_0)[\cdot]$ has a bounded Riesz representer $v^* \in \mathcal{V}$, so $v_{K_n}^* = \Pi_{K_n} v^* \rightarrow v^*$ and $\sigma_n \rightarrow \|v^*\|_{sd} < \infty$, delivering the parametric $\sqrt{n}$ rate. When Θ is irregular, no global representer exists and $\sigma_n \rightarrow \infty$, so the rate $\sigma_n/\sqrt{n}$ is slower than $1/\sqrt{n}$. The welfare and value functionals realize these two regimes, as we now show.

5.5 Application to the Welfare Functional

Take $\Theta = \overline{W}$. By the generalized Leibniz rule of Section 2.3, the boundary contribution to $D_\mu \overline{W}(\mu_0)[\cdot]$ vanishes, leaving the full-dimensional integral $D_\mu \overline{W}(\mu_0)[\overline{\psi}] = \int_{\{h_0 \geq 0\}} \overline{\psi}(x, \cdot) f(x) dx$, computed as in Section 3.3. The welfare functional is thus *regular*: its derivative admits the bounded Riesz representer

$$\nu^*(x, d) := \mathbb{1}\{h_0(x) \geq 0\} \lambda(x) \left(\frac{d}{p_0(x)} - \frac{1-d}{1-p_0(x)} \right)$$

of Theorem 1, so that $v_{K_n}^* = \Pi_{K_n} \nu^* \rightarrow \nu^*$ and $\sigma_n \rightarrow \sigma_W < \infty$. Theorem 7 then specializes as follows.

Corollary 1 (Debiased Machine Learning for the Welfare Functional). *Under the conditions of Theorem 7 with $\Theta = \overline{W}$, we have $\sigma_n \rightarrow \sigma_W < \infty$ and the cross-fitted sieve-debiased*

estimator (24) is $\sqrt{n}$ -asymptotically normal,

$$\sqrt{n} \left(\tilde{\theta} - W(h_0) \right) \xrightarrow{d} \mathcal{N} \left(0, \sigma_W^2 \right),$$

with σ_W^2 as in Theorem 1 in the known-density case, and with σ_W^2 replaced by $\bar{\sigma}_W^2$ of Theorem 2 (which adds the term $\text{Var} \left([h_0(X_i)]_+ \right)$) in the unknown-density case.

Remark 4 (Equivalent Closed-Form (AIPW) Construction). Because $\bar{W}$ is regular, its influence function ν^* is available in closed form, and the sieve-debiased estimator coincides asymptotically with the classical orthogonal-moment / DML construction of Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey and Robins (2018), which estimates the propensity score p_0 (and, when f_0 is unknown, the density ratio λ) directly rather than through the projection (23). Collecting the nuisances $\eta := (\mu, p, \lambda)$ and writing ν_η^* for ν^* evaluated at η , the orthogonal moment is

$$\psi(Z_i; \theta, \eta) := m(Z_i; \mu) - \theta + \nu_\eta^*(X_i, D_i)(Y_i - \mu(X_i, D_i)), \quad (25)$$

with $Z_i := (Y_i, D_i, X_i)$ and welfare term

$$m(Z_i; \mu) = \begin{cases} \int [\mu(x, 1) - \mu(x, 0)]_+ f(x) dx & (\text{known } f), \\ [\mu(X_i, 1) - \mu(X_i, 0)]_+ & (f = f_0, \lambda \equiv 1, \text{ AIPW score}). \end{cases}$$

This moment is Neyman-orthogonal at (θ_0, η_0) , and the corresponding cross-fitted estimator takes the explicit form

$$\hat{\theta}_{DML} := \frac{1}{n} \sum_{\kappa=1}^K \sum_{i \in I_\kappa} \left([\hat{\mu}_{-\kappa}(X_i, 1) - \hat{\mu}_{-\kappa}(X_i, 0)]_+ + \hat{\nu}_{-\kappa}^*(X_i, D_i)(Y_i - \hat{\mu}_{-\kappa}(X_i, D_i)) \right), \quad (26)$$

in the unknown-density case; in the known-density case the per-observation welfare term $[\hat{\mu}_{-\kappa}(X_i, 1) - \hat{\mu}_{-\kappa}(X_i, 0)]_+$ is replaced by the fold-level deterministic integral $\bar{W}(\hat{\mu}_{-\kappa})$, so that (with equal-sized folds) $\hat{\theta}_{DML} = \frac{1}{K} \sum_{\kappa} \bar{W}(\hat{\mu}_{-\kappa}) + \frac{1}{n} \sum_{\kappa} \sum_{i \in I_\kappa} \hat{\nu}_{-\kappa}^*(X_i, D_i)(Y_i - \hat{\mu}_{-\kappa}(X_i, D_i))$. It attains the same limit as Corollary 1 under, for each fold κ : (i) strict overlap, $\hat{p}_{-\kappa} \in [\underline{p}, \bar{p}] \subset (0, 1)$ with probability approaching one; (ii) consistency, $\|\hat{\mu}_{-\kappa} - \mu_0\|_\infty = o_p(1)$ and $\|\hat{p}_{-\kappa} - p_0\|_\infty = o_p(1)$ (and $\|\hat{\lambda}_{-\kappa} - \lambda\|_\infty = o_p(1)$ when f_0 is unknown); and (iii) the product rate $\|\hat{\mu}_{-\kappa} - \mu_0\|_2 \left(\|\hat{p}_{-\kappa} - p_0\|_2 + \|\hat{\lambda}_{-\kappa} - \lambda\|_2 \right) + \|\hat{\mu}_{-\kappa} - \mu_0\|_\infty^2 = o_p(n^{-1/2})$. This construction dispenses with the Donsker condition of Assumption 3(c), and its proof is given in Appendix A. The variance is then consistently estimated by the empirical score variance $\hat{\sigma}^2 := \frac{1}{n} \sum_{\kappa} \sum_{i \in I_\kappa} \hat{\psi}_i^2$ with $\hat{\psi}_i := \psi(Z_i; \hat{\theta}_{DML}, \hat{\eta}_{-\kappa})$. Simulation evidence for a cross-fitted welfare estimator is reported in Appendix C.4.

5.6 Application to the Value Functional

Take $\Theta = \bar{V}$ under a known target density f . As established in Section 2.3, the pathwise derivative $D_\mu \bar{V}(\mu_0)[\cdot]$ is a submanifold integral of dimension $m = d_x - 1$ and is unbounded on $L^2(f)$, so no influence function exists on the full Hilbert space and the standard DML recipe is unavailable. The sieve Riesz representer (23), however, remains well-defined and supplies the local influence function, with $\hat{D}_\mu \bar{V}(\hat{\mu})[\bar{\psi}]$ the numerical (ϵ -band) pathwise derivative of Section 4.1. Because the value functional is irregular, the sieve Riesz norm of Theorem 7 diverges: $\sigma_n = \sigma_{V,n}$ with $\sigma_{V,n}^2 := \|v_{K_n}^*\|_{sd}^2 \asymp K_n^{1/d_x} \rightarrow \infty$ (so $\sigma_{V,n} \asymp K_n^{1/(2d_x)}$), matching the sieve variance of Theorem 3.

Corollary 2 (Sieve-Debiased Estimation of the Value Functional). *Under the conditions of Theorem 7 with $\Theta = \bar{V}$, and with $\sigma_{V,n}^2 = \|v_{K_n}^*\|_{sd}^2 \asymp K_n^{1/d_x}$ as in Theorem 3, the cross-fitted sieve-debiased estimator (24) is asymptotically normal,*

$$\frac{\sqrt{n}(\tilde{\theta} - \theta_0)}{\sigma_{V,n}} \xrightarrow{d} \mathcal{N}(0, 1),$$

and the sieve variance estimator $\hat{\sigma}_V^2$ of Section 4.1 remains consistent, so that the studentized confidence intervals of Section 4.1 apply with $\tilde{\theta}$ in place of $\bar{V}(\hat{\mu})$.

Remark 5 (Weaker Rate Requirements than Regular DML). The first-stage rate requirements for the irregular value functional are in fact *weaker* than in regular DML. Assumption 5(c) requires only $\|\hat{v}_{K_n, -\kappa}^* - v_{K_n}^*\| \cdot \|\hat{\mu}_{-\kappa} - \mu_0\| = o_p\left(\sqrt{K_n^{1/d_x}/n}\right)$, whereas in regular locally robust estimation — where the Riesz representer is well-defined on the whole space and the $\sqrt{n}$ rate is attainable — the corresponding requirement is the stronger $\|\hat{v}^* - v^*\| \cdot \|\hat{\mu} - \mu_0\| = o_p(1/\sqrt{n})$ (e.g., Chernozhukov *et al.*, 2018, 2022). Since $K_n^{1/d_x} \rightarrow \infty$, the irregularity of the value functional, while slowing the attainable convergence rate, also relaxes the accuracy demanded of the first-stage estimators. The comparison is between the *tolerances* required of the same product $\|\hat{v}^* - v^*\| \cdot \|\hat{\mu} - \mu_0\|$: the irregular case permits a slower-vanishing bound $\sqrt{K_n^{1/d_x}/n} \gg n^{-1/2}$, reflecting that the target itself converges only at the slower rate.

Remark 6 (Generic Machine-Learning First Stages). Unlike the plug-in Theorem 3, the sieve-debiased estimator does not require $\hat{\mu}$ to be the stacked sieve least squares estimator from (17): Assumption 5 imposes only high-level rate conditions, which generic machine-learning first-stage learners may satisfy. The sieve structure is retained only in the construction of the local influence function $\hat{v}_{K_n}^*$, which is computable in closed form from (23) regardless of how $\hat{\mu}$ is obtained. For generic first stages, Assumption 5(d) requires the sieve space $\mathcal{V}_{K_n}$ to track the first-stage estimation error sufficiently well, in analogy to the Riesz-representer approximation conditions in the automatic debiased machine learning literature.

Remark 7 (Comparison with the Split-Sample and Leave-One-Out Debiasing). The debiasing device here and the quadratic debiasing of Section 4.4 target different terms in the expansion of the plug-in estimator. The SS and LOO estimators of Section 4.4 retain the sieve least squares first stage and remove the diagonal part of the *quadratic* remainder, thereby weakening the smoothness requirement on h_0 from $s > d_x$ to $s > (d_x + 1)/2$. The sieve-debiased estimator $\tilde{\theta}$ instead annihilates the *first-order* term on the sieve space, thereby accommodating generic machine-learning first stages. Whether the two devices can be fruitfully combined — a cross-fitted sieve influence-function correction together with a diagonal quadratic correction, so as to simultaneously weaken the smoothness requirement and accommodate machine-learning first stages — is an interesting question that we leave for future research.

6 Simulations

6.1 Results for Theorem 1

We first report the finite-sample performance of the semiparametric estimator, its associated standard error estimator, and the resulting confidence interval, based on the theoretical results in Theorem 1, which concern the welfare functional under a known target distribution. The model specifications used in the simulations are summarized in Table 1. For each specification, random samples of size n are drawn with covariates $X_i \sim F_0$, and treatment status is assigned according to the propensity score function $p_0(X_i)$. Outcomes are then generated as $Y_i = \mu_0(X_i, D_i) + \epsilon_i$, with $\epsilon_i \sim N(0, 1)$.

Table 1: Theorem 1: Model Specifications

Model	F_0	F	$\mu_0(x, d)$	$p_0(x)$	σ^2
M1	$U[-0.2, 1.2]$	$U[0, 1]$	$5 \sin(2\pi x) \cos(2\pi x) + d(-0.4 + 2x^2)$	$\frac{1}{1+e^{-(1-2x)}}$	1
M2	$U[-0.2, 1.2]$	$U[0, 1]$	$0.5 x + d(0.5 - x^2)$	$\frac{1}{1+e^{-(0.5+x)}}$	1
M3	$U[-0.2, 1.2]$	$U[0, 1]$	$x^2 + d(1 - x)$	$\frac{1}{1+e^{-(0.5-x)}}$	1
M4	$U[-0.2, 1.2]^2$	$U[0, 1]^2$	$(1 - x_1^2 - x_2^2)(4 + \sin x_1 x_2 + \cos x_2) + d(0.5x_1 - 0.4x_2)$	$\frac{1}{1+e^{-(x_1-x_2)}}$	1
M5	$U[-0.2, 1.2]^2$	$U[0, 1]^2$	$(1 - x_1 x_2)(3 + \sin(\pi x_1) \cos(\pi x_2)) + d(0.3x_1 - 0.3x_2)$	$\frac{1}{1+e^{-(x_1-x_2)}}$	1
M6	$U[-0.2, 1.2]^2$	$U[0, 1]^2$	$\log(1 + x_1 + x_2) + d(x_1 - 0.7x_2)$	$\frac{1}{1+e^{-(1.5x_1-0.5x_2)}}$	1
M7	$U[-0.2, 1.2]^2$	$U[0, 1]^2$	$(x_1^2 + x_2^2)e^{-(x_1+x_2)} + d(0.5 - x_2)$	$\frac{1}{1+e^{-(0.5+x_1+2x_2)}}$	1

The semiparametric estimator of the welfare functional is constructed in two steps. In the first step, $\mu_0(x, 1)$ and $\mu_0(x, 0)$ are estimated separately for the treated group ($D_i = 1$) and the control group ($D_i = 0$) using B-spline regressions. In the second step, the welfare functional is approximated by numerically integrating over $M = 5,000$ Sobol points $\{X_j^{Sobol}\}_{j=1}^M$ drawn from the target distribution F :

$$\widehat{W}(\hat{\mu}) = \frac{1}{M} \sum_{j=1}^M [\hat{\mu}(X_j^{Sobol}, 1) - \hat{\mu}(X_j^{Sobol}, 0)]_+,$$

where $\hat{\mu}(x, 1)$ and $\hat{\mu}(x, 0)$ denote the first-stage nonparametric estimators.

Let $\hat{h}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$ denote the estimated CATE function, and let $\hat{p}(x)$ be a B-spline sieve estimator of the propensity score. The 95% confidence interval for the welfare functional takes the usual form,

$$\left[\widehat{W}(\hat{\mu}) - 1.96 \frac{\hat{\sigma}_W}{\sqrt{n}}, \widehat{W}(\hat{\mu}) + 1.96 \frac{\hat{\sigma}_W}{\sqrt{n}} \right],$$

with

$$\hat{\sigma}_W^2 = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\hat{h}(X_i) \geq 0\} \lambda^2(X_i) (Y_i - \hat{\mu}(X_i, D_i))^2}{\hat{p}(X_i) (1 - \hat{p}(X_i))}.$$

To evaluate performance, we simulate each model 2,000 times at sample sizes $n = 1500, 3000$, and 6000 . Table 2 reports the true welfare functional (W_{true}), the average bias of the estimator $\widehat{W}(\hat{\mu})$ (Bias), its sampling standard deviation (SD), the average estimated standard error (SE), the standard deviation of SE across iterations (SD(SE)), and the empirical coverage probability of the associated 95% confidence interval (Coverage). The results³ indicate that the nominal coverage rate is attained in nearly all cases, even with relatively small samples, and improves further as sample size increases. The bias of the estimator also becomes negligible relative to its sampling variability, and the proposed SE estimator closely tracks the sampling standard deviation with high precision.

Remark 1. *To improve computational efficiency, we predetermine the sieve dimensions for estimating $\mu_0(x, 0)$ and $\mu_0(x, 1)$. For each model specification, we first generate a dataset $(Y_i, X_i, D_i)_{i=1}^{n=6000}$ and apply the sieve dimension selection procedure of [Chen, Christensen and Kankanala \(2025\)](#) separately to the treated and control groups. As demonstrated in their paper, this approach ensures that the resulting estimators of $\mu_0(x, 1)$ and $\mu_0(x, 0)$ converge at the fastest possible (i.e., minimax) rates in the sup-norm. The selected sieve dimensions are then used in the simulation designs with $n = 1500, 3000, 6000$. In principle, one could implement data-driven dimension selection within each simulation iteration, but doing so would substantially increase computational cost.*

³Additional simulations based on GAM are presented in Appendix C.1.

Table 2: Theorem 1: Simulation Results

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M1	1500	0.3857	0.0161	0.0470	0.0480	0.0033	0.9415
	3000	0.3857	0.0080	0.0338	0.0337	0.0018	0.9480
	6000	0.3857	0.0037	0.0229	0.0237	0.0010	0.9535
M2	1500	0.2358	0.0030	0.0484	0.0518	0.0031	0.9615
	3000	0.2358	0.0028	0.0352	0.0366	0.0015	0.9555
	6000	0.2358	0.0009	0.0243	0.0259	0.0007	0.9630
M3	1500	0.5001	0.0046	0.0579	0.0605	0.0025	0.9620
	3000	0.5001	0.0025	0.0407	0.0430	0.0013	0.9595
	6000	0.5001	0.0000	0.0293	0.0305	0.0006	0.9625
M4	1500	0.1033	0.0185	0.0478	0.0561	0.0074	0.9735
	3000	0.1033	0.0088	0.0348	0.0400	0.0040	0.9745
	6000	0.1033	0.0043	0.0248	0.0284	0.0021	0.9695
M5	1500	0.0499	0.0282	0.0418	0.0521	0.0093	0.9700
	3000	0.0499	0.0158	0.0303	0.0369	0.0056	0.9720
	6000	0.0499	0.0094	0.0227	0.0262	0.0032	0.9660
M6	1500	0.2315	0.0268	0.0559	0.0622	0.0048	0.9550
	3000	0.2315	0.0123	0.0394	0.0444	0.0025	0.9655
	6000	0.2315	0.0050	0.0288	0.0315	0.0014	0.9695
M7	1500	0.1250	0.0462	0.0505	0.0595	0.0073	0.9370
	3000	0.1250	0.0218	0.0346	0.0406	0.0043	0.9610
	6000	0.1250	0.0101	0.0245	0.0278	0.0022	0.9680

Notes: (1) W_{true} denotes the true welfare functional, computed numerically using 5,000 Sobol points drawn from F . (2) Bias is the average deviation from W_{true} . (3) SD is the sampling standard deviation. (4) SE is the average estimated asymptotic standard error of $\hat{W}(\hat{\mu})$. (5) SD(SE) is the standard deviation of SE across iterations. (6) Coverage is the empirical coverage probability of the 95% CI.

Remark 2. The construction of the 95% confidence interval requires estimating the propensity score function $p_0(x)$. To this end, we again use the B-spline sieve estimator, regressing treatment status on the covariates. The sieve dimension is similarly predetermined following [Chen, Christensen and Kankanala \(2025\)](#), based on the full dataset. A potential concern is that the fitted propensity score $\hat{p}(x)$ may take values outside the unit interval for some observations, which is likely to indicate a violation of the strict overlap assumption. Accordingly, when computing the asymptotic standard deviation $\hat{\sigma}_W$, we trim observations with estimated propensity scores lying outside $[0, 1]$.

6.2 Results for Theorem 2

We now investigate the finite-sample performance of our estimation and inference procedure for the welfare functional in the case where the target distribution F coincides with the population distribution F_0 , though both remain unknown. The relevant model specifications are presented in Table 3. In contrast to the designs considered in Section 6.1, these specifications explicitly impose that F and F_0 share identical supports.

Table 3: Theorem 2: Model Specifications

Model	F_0	F	$\mu_0(x, d)$	$p_0(x)$	σ^2
M8	U[0, 1]	U[0, 1]	$5 \sin(2\pi x) \cos(2\pi x) + d(-0.4 + 2x^2)$	$\frac{1}{1+e^{-(1-2x)}}$	1
M9	U[0, 1]	U[0, 1]	$0.5 x + d(0.5 - x^2)$	$\frac{1}{1+e^{-(-0.5+x)}}$	1
M10	U[0, 1]	U[0, 1]	$x^2 + d(1 - x)$	$\frac{1}{1+e^{-(0.5-x)}}$	1
M11	U[0, 1] ²	U[0, 1] ²	$(1 - x_1^2 - x_2^2)(4 + \sin x_1 x_2 + \cos x_2) + d(0.5x_1 - 0.4x_2)$	$\frac{1}{1+e^{-(x_1-x_2)}}$	1
M12	U[0, 1] ²	U[0, 1] ²	$(1 - x_1 x_2)(3 + \sin(\pi x_1) \cos(\pi x_2)) + d(0.3x_1 - 0.3x_2)$	$\frac{1}{1+e^{-(x_1-x_2)}}$	1
M13	U[0, 1] ²	U[0, 1] ²	$\log(1 + x_1 + x_2) + d(x_1 - 0.7x_2)$	$\frac{1}{1+e^{-(1.5x_1-0.5x_2)}}$	1
M14	U[0, 1] ²	U[0, 1] ²	$(x_1^2 + x_2^2)e^{-(x_1+x_2)} + d(0.5 - x_2)$	$\frac{1}{1+e^{-(-0.5+x_1+2x_2)}}$	1

The plug-in estimator $\hat{\bar{W}}(\hat{\mu})$ proposed here differs from that in Section 6.1 only in the second step: instead of using Sobol points to approximate the integral, we take the sample average of $[\hat{\mu}(X_i, 1) - \hat{\mu}(X_i, 0)]_+$ over the observed data:

$$\hat{\bar{W}}(\hat{\mu}) = \frac{1}{n} \sum_{i=1}^n [\hat{\mu}(X_i, 1) - \hat{\mu}(X_i, 0)]_+.$$

Relative to the known- F case, the asymptotic variance of $\hat{\bar{W}}(\hat{h})$ includes an additional component, $\text{Var}([h_0(X_i)]_+)$. We estimate the asymptotic standard deviation $\hat{\sigma}_W$ by substi-

tuting the B-spline sieve estimates for the nuisance functions and replacing the population mean with its sample analog. As before, we restrict attention to observations with estimated propensity scores in $[0, 1]$. A 95% confidence interval is then given by

$$\left[\hat{W}(\hat{\mu}) - 1.96 \frac{\hat{\sigma}_W}{\sqrt{n}}, \quad \hat{W}(\hat{\mu}) + 1.96 \frac{\hat{\sigma}_W}{\sqrt{n}} \right].$$

The simulation results⁴ based on 2,000 iterations with sample sizes $n = 1500, 3000$, and 6000 are reported in Table 4. Overall, the coverage of the proposed confidence intervals converges to the nominal 95% level as sample size increases, while the decreasing bias and standard deviation indicate a reduction in mean squared error.

Remark 3. *Extrapolation bias may arise when B-spline fits are evaluated outside the support of their training samples—for instance, when $\hat{\mu}(x, 1)$ is evaluated using control group data or $\hat{\mu}(x, 0)$ using treated group data. To mitigate this issue, we trim observations that fall outside the common support of treated and control groups, estimate the nuisance functions on the trimmed sample, and then compute the welfare functional. Since only a small fraction of observations are removed, this adjustment has a negligible effect on the results.*

6.3 Results for Theorem 3

To assess the finite-sample properties of our estimation and inference procedure for the value functional $\bar{V}(\hat{\mu})$ under a known target distribution F , we analyze the model described in Table 5:

Table 5: Theorem 3: Model Specification

Model	F_0	F	$\mu_0(x_1, x_2, d)$	$v_0(x_1, x_2)$	$p_0(x_1, x_2)$	σ^2
M15	$U([-2, 2]^2)$	$U([-1.5, 1.5]^2)$	$d \cdot (1 - x_1^2 - x_2^2) \cdot (4 + \sin(x_1)x_2 + \cos(x_2))$	1	$\frac{1}{1+e^{-(x_1-x_2)}}$	1

Our parameter of interest is a scaled value functional under known F :

$$V(h_0) = 3^2 \int \mathbb{1} \left\{ \left(1 - x_1^2 - x_2^2 \right) \left(4 + \sin(x_1)x_2 + \cos(x_2) \right) \geq 0 \right\} dF(x_1, x_2),$$

where the scaling factor 3^2 is chosen so that the integral evaluates to π . Equivalently, this setup can be interpreted as a Monte Carlo experiment for estimating the area of the unit circle by uniformly throwing darts over a 3×3 square.

The plug-in estimator $\hat{\bar{V}}(\hat{\mu})$ for $\bar{V}(\mu_0)$ is constructed in two steps. In the first step, we estimate $\mu_0(x, 1) = \mu_0(x_1, x_2, 1)$ and $\mu_0(x, 0) = \mu_0(x_1, x_2, 0)$ separately using B-spline sieve

⁴Additional simulations based on GAM are presented in Appendix C.2.

Table 4: Theorem 2: Simulation Results

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M8	1500	0.3857	0.0068	0.0414	0.0423	0.0025	0.9515
	3000	0.3857	0.0029	0.0296	0.0297	0.0013	0.9550
	6000	0.3857	0.0006	0.0206	0.0210	0.0007	0.9575
M9	1500	0.2358	0.0042	0.0425	0.0440	0.0025	0.9605
	3000	0.2358	0.0029	0.0304	0.0311	0.0012	0.9590
	6000	0.2358	0.0012	0.0209	0.0221	0.0006	0.9555
M10	1500	0.5001	0.0067	0.0511	0.0515	0.0018	0.9495
	3000	0.5001	0.0037	0.0357	0.0366	0.0009	0.9600
	6000	0.5001	0.0013	0.0253	0.0260	0.0005	0.9600
M11	1500	0.1033	0.0307	0.0365	0.0402	0.0038	0.9245
	3000	0.1033	0.0156	0.0264	0.0286	0.0021	0.9395
	6000	0.1033	0.0084	0.0192	0.0203	0.0012	0.9470
M12	1500	0.0499	0.0418	0.0316	0.0373	0.0045	0.8790
	3000	0.0499	0.0232	0.0229	0.0263	0.0029	0.9135
	6000	0.0499	0.0126	0.0168	0.0186	0.0017	0.9355
M13	1500	0.2315	0.0177	0.0432	0.0459	0.0032	0.9565
	3000	0.2315	0.0088	0.0309	0.0323	0.0014	0.9510
	6000	0.2315	0.0041	0.0221	0.0229	0.0007	0.9525
M14	1500	0.1250	0.0251	0.0382	0.0421	0.0059	0.9460
	3000	0.1250	0.0117	0.0258	0.0287	0.0030	0.9585
	6000	0.1250	0.0054	0.0182	0.0198	0.0013	0.9595

Notes: (1) W_{true} is the true welfare functional, computed numerically using 5,000 Sobol points drawn from F . (2) Bias is the average deviation from W_{true} . (3) SD is the sampling standard deviation. (4) SE is the average estimated asymptotic standard error of $\hat{W}(\hat{\mu})$. (5) SD(SE) is the standard deviation of SE across iterations. (6) Coverage is the empirical coverage probability of the 95% CI.

estimators, fitted on the treated and control groups, respectively, with sieve dimensions predetermined as described in subsection 6.1. In the second step, $\mathbb{1}\{\hat{\mu}(x, 1) - \hat{\mu}(x, 0) \geq 0\}$ is numerically integrated using 5,000 Sobol points drawn from F . The resulting semiparametric two-step estimator is

$$\hat{V}(\hat{\mu}) = \frac{1}{M} \sum_{j=1}^M \mathbb{1}\{\hat{\mu}(X_j^{Sobol}, 1) - \hat{\mu}(X_j^{Sobol}, 0) \geq 0\}.$$

The 95% confidence interval is then constructed as

$$\left[\hat{V}(\hat{\mu}) - 1.96 \frac{\hat{\sigma}_V}{\sqrt{n}}, \quad \hat{V}(\hat{\mu}) + 1.96 \frac{\hat{\sigma}_V}{\sqrt{n}} \right],$$

where the asymptotic variance estimate $\hat{\sigma}_V^2$ is given by

$$\hat{\sigma}_V^2 = \hat{D}_\mu \bar{V}(\hat{\mu}) [\bar{\psi}]' \hat{\Omega} \hat{D}_\mu \bar{V}(\hat{\mu}) [\bar{\psi}],$$

with $\hat{\Omega}$ being the estimated asymptotic covariance matrix for the OLS estimators in the linear regression model (17), and

$$\hat{D}_\mu \bar{V}(\hat{\mu}) [\bar{\psi}] = 3^2 \left(\frac{1}{2\epsilon} \int_{\{x \in [-1.5, 1.5]^2 : -\epsilon < \hat{h}(x) < \epsilon\}} \psi^{(K_1)}(x) \frac{1}{3^2} dx - \frac{1}{2\epsilon} \int_{\{x \in [-1.5, 1.5]^2 : -\epsilon < \hat{h}(x) < \epsilon\}} \psi^{(K_0)}(x) \frac{1}{3^2} dx \right).$$

We set the tuning parameter $\epsilon = 0.005$ to mitigate the approximation bias in $\hat{D}_\mu \bar{V}(\hat{\mu})[\bar{\psi}]$, which we compute using the sample average over M Sobol draws from F . Because draws from F are unlikely to fall within the set $\{x \in [-1.5, 1.5]^2 : -\epsilon < \hat{h}(x) < \epsilon\}$ when ϵ is small, we use $M = 1,000,000$ Sobol points to ensure accuracy. Simulation results⁵ based on 2,000 iterations with sample sizes $n = 1500, 3000$, and 6000 are reported in Table 6. The results show that the coverage rate reaches the nominal level with relatively modest sample sizes, even though the value functional is not $\sqrt{n}$ -estimable. Also, as sample size increases, both bias and standard error decrease, and the plug-in estimator for the standard error provides a close estimate of the sampling standard deviation.

6.4 Results for Section 5: DML via the Sieve Riesz Representer

We now investigate the unified procedure of Section 5 numerically, with a *generic* machine-learning first stage. The data-generating process draws $X_i \sim U[0, 1]^2$ and $Y_i = \mu_0(X_i, 0) + D_i h_0(X_i) + \epsilon_i$ with $\epsilon_i \sim N(0, 1)$; writing $a_1 = (1, -1)/\sqrt{2}$, $a_2 = (1, 1)/\sqrt{2}$ and $\bar{a} = \sqrt{2}/2$, the CATE is the smooth two-index ridge function $h_0(x) = 3[\Lambda(3a_1'x) + \frac{1}{2}\Lambda(3(a_2'x - \bar{a})) - 0.70]$ (Λ the logistic c.d.f.), with baseline $\mu_0(x, 0) = 2 \tanh(2(a_2'x - \bar{a})) + \sin(2\pi a_1'x)$ and propensity

⁵Additional simulations under alternative model specifications, as well as robustness checks for different values of ϵ , are presented in Appendix C.3.

Table 6: Theorem 3: Simulation Results

Model	n	V_{true}	Bias	SD	SE	SD(SE)	Coverage
M15	1500	3.1416	0.0076	0.0710	0.0711	0.0092	0.9420
	3000	3.1416	0.0080	0.0486	0.0499	0.0029	0.9475
	6000	3.1416	0.0062	0.0337	0.0353	0.0016	0.9490

Notes: (1) V_{true} is the true value functional, equal to π . (2) Bias is the average deviation from V_{true} . (3) SD is the sampling standard deviation of the estimator. (4) SE is the average estimated asymptotic standard error across iterations. (5) SD(SE) is the standard deviation of SE across iterations. (6) Coverage is the empirical coverage probability of the 95% confidence interval.

$p_0(x) = \Lambda(1.2a'_1x + 0.4(a'_2x - \bar{a}))$. Under $F = \text{U}[0, 1]^2$ this gives a regular welfare functional $W(h_0) = 0.3555$ and an irregular value functional $V(h_0) = 0.5681$. Crucially, and unlike the sieve plug-in of Sections 3–4, the first stage $\hat{\mu}_{-\kappa}(\cdot, d)$ is a *generic* machine learner, not the sieve least squares estimator — the regime targeted by Theorem 7. To probe robustness to the choice of learner, we report two first stages of very different character: per-arm *gradient boosting* (?) (tree ensembles) and a per-arm *random-feature network* (a two-layer net with a random Fourier-feature hidden layer (?) and a ridge output layer tuned by leave-one-out cross-validation). In both cases the sieve Riesz representer $\hat{v}_{K_n}^*$ of (23) is built on the *same* tensor cubic B-spline sieve ($K_n = 81$, with a small ridge stabilizing the collinear Gram), the estimator $\tilde{\theta}$ of (24) is cross-fitted with $\mathcal{K} = 5$ folds, and inference uses the studentized interval $\tilde{\theta} \pm 1.96 \hat{\sigma}_n / \sqrt{n}$ with the internally consistent sieve-Riesz variance $\hat{\sigma}_n^2 = \frac{1}{n} \sum_i \hat{v}_{K_n, -\kappa(i)}^*(X_i, D_i)^2 (Y_i - \hat{\mu}_{-\kappa(i)}(X_i, D_i))^2$. The two learners are run on identical samples and cross-fitting folds (a paired comparison); Table 7 reports 250 Monte-Carlo iterations. (An earlier, welfare-only cross-fitted experiment with a generalized additive model first stage — a precursor specialized to the regular case — is reported in Appendix C.4.)

The results confirm Corollaries 1–2 and, importantly, are robust across the two first stages. For the regular welfare functional, the boosting plug-in carries a sizeable upward bias (the curvature of $[\cdot]_+$) that the sieve-Riesz correction removes almost entirely, with the standard deviation declining at the parametric $\sqrt{n}$ rate, the studentized standard error tracking it closely, and coverage at the nominal level — all with a generic first stage and no closed-form influence function. The random-feature network behaves differently in an instructive way: being a smooth global fit, its plug-in is already nearly unbiased, so the correction is less consequential and, at small n , introduces a finite-sample bias (the second-order remainder of Assumption 5(d)–(e), which the sieve only partially absorbs for a non-sieve learner); this bias decays as n grows ($-0.033 \rightarrow -0.011 \rightarrow -0.006$) and coverage rises to the nominal level ($0.91 \rightarrow 0.92 \rightarrow 0.96$), exactly as the theory predicts. For the irregular

Table 7: Section 5: DML via the Sieve Riesz Representer under two generic first stages ($\mathcal{K} = 5$ cross-fitting, $K_n = 81$, 250 iterations). Truths $W(h_0) = 0.3555$, $V(h_0) = 0.5681$.

Functional	n	Plug-in bias	DML bias	DML SD	SE	Coverage
<i>Panel A: gradient-boosting first stage</i>						
Welfare W	1500	0.0287	−0.0030	0.0514	0.0531	0.960
	3000	0.0216	0.0032	0.0305	0.0308	0.964
	6000	0.0127	0.0014	0.0209	0.0203	0.936
Value V	1500	0.0008	0.0018	0.0839	0.0652	0.900
	3000	0.0058	0.0024	0.0420	0.0351	0.896
	6000	0.0022	−0.0027	0.0274	0.0242	0.900
<i>Panel B: random-feature network first stage</i>						
Welfare W	1500	0.0083	−0.0332	0.0628	0.0561	0.912
	3000	0.0052	−0.0111	0.0312	0.0314	0.924
	6000	0.0010	−0.0057	0.0210	0.0205	0.956
Value V	1500	0.0086	−0.0155	0.0974	0.0813	0.908
	3000	0.0071	0.0007	0.0512	0.0482	0.924
	6000	0.0000	−0.0041	0.0315	0.0312	0.928

Notes: “Plug-in bias” is the average deviation of the uncorrected machine-learning plug-in $\Theta(\hat{\mu})$; “DML bias”, “DML SD”, “SE”, and “Coverage” refer to the cross-fitted sieve-debiased estimator $\hat{\theta}$, its sampling standard deviation, the average studentized standard error $\hat{\sigma}_n/\sqrt{n}$, and the empirical 95% coverage, over 250 iterations on identical samples. Both first stages use the same $K_n = 81$ tensor cubic B-spline sieve for the Riesz representer.

value functional, both learners yield an essentially centered estimator with close-to-nominal coverage (≈ 0.90 – 0.93) despite the absence of $\sqrt{n}$ -estimability; the mild anti-conservatism reflects the second-order remainder that limits the rate. In short, the qualitative conclusions of Section 5 — bias removal, valid studentized inference, and the regular/irregular distinction — hold under both machine-learning first stages.

The defining distinction between the two regimes of Theorem 7 is the behavior of the sieve-Riesz norm $\sigma_n = \|v_{K_n}^*\|_{sd}$: bounded for a regular functional, divergent for an irregular one. Table 8 isolates this in the cleanest setting, $d_x = 1$, where the irregular norm grows like $\sigma_n \asymp K_n^{1/(2d_x)} = \sqrt{K_n}$. Sweeping the B-spline dimension K_n at fixed $n = 6000$, the welfare norm $\hat{\sigma}_n^W$ is essentially flat, whereas the value norm $\hat{\sigma}_n^V$ rises steadily and tracks the $\sqrt{K_n}$ growth rate (up to a constant; the reference row in Table 8 reports the unnormalized $\sqrt{K_n}$, so only the growth shape, not the level, should be compared) — the numerical signature of the regular/irregular dichotomy, and the mechanism by which the same estimator delivers the $\sqrt{n}$ rate for W and the slower minimax rate for V .

Table 8: Sieve-Riesz norm $\hat{\sigma}_n = \|\hat{v}_{K_n}^*\|_{sd}$ versus sieve dimension ($d_x = 1$, $n = 6000$): bounded for the welfare functional, $\propto \sqrt{K_n}$ for the value functional

K_n	6	10	16	24	36	52
$\hat{\sigma}_n^W$ (welfare, regular)	1.47	1.50	1.50	1.52	1.54	1.57
$\hat{\sigma}_n^V$ (value, irregular)	1.13	1.56	1.87	2.07	2.41	2.64
$\sqrt{K_n}$ (reference)	2.45	3.16	4.00	4.90	6.00	7.21

Moderate-to-high-dimensional study. Tables 9–?? extend the experiment to $d_x \in \{10, 50, 100\}$. The covariates are $X \sim \text{U}[0, 1]^{d_x}$ and the CATE is the same two-index ridge $h_0(x) = 3[\Lambda(3(a_1'x - m_1)) + \frac{1}{2}\Lambda(3(a_2'x - m_2)) - 0.70]$ with *dense* unit indices $a_1 = (+1, -1, +1, \dots)/\sqrt{d_x}$, $a_2 = \mathbf{1}/\sqrt{d_x}$ and midpoints $m_j = \frac{1}{2} \sum_k (a_j)_k$, so that only two linear indices carry signal among the d_x covariates and the welfare/value truths stay d_x -stable ($W(h_0) \approx 0.34$, $V(h_0) \approx 0.585$) while the first-stage regression is a genuine high-dimensional learning problem (find a two-dimensional structure in noise). Because a tensor B-spline sieve is infeasible here—its dimension $(n_{\text{knots}} + \text{deg} - 1)^{d_x}$ explodes—the Riesz representer is built on a fixed-dimension *random-feature* (cosine) sieve, $\bar{\psi}(w)$ stacking $K_n = 300$ features $\cos(\gamma \omega_k' w + b_k)$ with random (ω_k, b_k) and a frequency scale γ ; everything else—the generic per-arm gradient-boosting first stage, $\mathcal{K} = 5$ cross-fitting, and the studentized interval $\tilde{\theta} \pm 1.96 \hat{\sigma}_n/\sqrt{n}$ —is exactly as in the $d_x = 2$ study.

Two messages emerge. First, the estimator continues to deliver bias removal and valid studentized inference well into high dimension: across all nine (d_x, n) cells the plug-in bias—

sizeable, and for the value functional growing with d_x —is removed to within Monte-Carlo error, the DML standard deviation falls at the $\sqrt{n}$ rate, the studentized standard error tracks it, and coverage is close to nominal (Welfare 0.92–0.98, Value 0.91–0.96). Second, high dimension makes the sieve resolution γ —the random-feature analogue of the sieve dimension K_n , i.e. an undersmoothing parameter—both consequential and *functional-specific* (Table ??). The two functionals pull in opposite directions. The regular welfare functional, whose Riesz representer is a bulk object, requires a *richer* (higher- γ) sieve as d_x grows to keep the sieve-approximation bias negligible (Assumption 5(d)); too smooth a sieve leaves an $O(1)$ bias that, fatally, becomes visible only as n grows—at $d_x = 100$ the $\gamma = 3$ welfare coverage falls from 0.88 to 0.65 as $n : 4000 \rightarrow 16000$ while the standard error shrinks. The irregular value functional, whose representer concentrates on the decision boundary $\{h_0 = 0\}$, requires a *smoother* (lower- γ) sieve: in high dimension a high-frequency random feature oscillates predominantly along the $d_x - 2$ nuisance directions—a random unit direction places only $O(1/d_x)$ of its energy on the two-index signal plane—which ill-conditions the Gram and inflates the representer, collapsing coverage for $\gamma \geq 5$ once $d_x \geq 50$. Accordingly we set $\gamma_W = 3, 3, 5$ and $\gamma_V = 3, 2, 1.5$ at $d_x = 10, 50, 100$ (Table 9). A first-stage lever acts in the same functional-specific way: the welfare functional, being convex, favors a lower-variance (regularized) learner to control its second-order remainder (Assumption 5(e)), whereas the value functional favors a lower-bias (early-stopped, capacity-adaptive) learner so that its plug-in boundary bias vanishes with n rather than growing; the configurations reported pair each functional with the first stage it prefers.

7 Empirical Application

We revisit the empirical application analyzed in Kitagawa and Tetenov (2018) using the Job Training Partnership Act (JTPA) dataset. The JTPA study randomized whether applicants were eligible to receive a mix of training, job-search assistance, and other services provided under the program for a period of 18 months. A detailed description of the study and an assessment of average program effects for five major subgroups of the target population are provided in Bloom, Orr, Bell, Cave, Doolittle, Lin and Bos (1997).

We evaluate welfare using two outcome measures, following the approach in Kitagawa and Tetenov (2018). The first is total earnings over the 30 months following treatment assignment. The second adjusts this measure by subtracting \$774 for individuals assigned to treatment, thereby incorporating program costs. These outcomes are considered from an intention-to-treat perspective, meaning we focus on eligibility assignment rather than treatment effects among compliers. The available covariates include applicants’ pre-program

Table 9: Section 5: DML via the Sieve Riesz Representer in moderate-to-high dimension ($d_x \in \{10, 50, 100\}$), random-feature cosine Riesz sieve ($K_n = 300$), $\mathcal{K} = 5$ cross-fitting, 300 iterations. Functional-specific configuration: Welfare uses a regularized gradient-boosting first stage and sieve frequency γ_W ; Value uses an early-stopping gradient-boosting first stage and frequency γ_V (see text).

d_x (γ)	n	Plug-in bias	DML bias	DML SD	SE	Coverage
<i>Panel A: Welfare W (regularized boosting, γ_W)</i>						
$d_x = 10$ ($\gamma = 3$)	4000	0.0579	−0.0040	0.0309	0.0381	0.983
	8000	0.0430	0.0088	0.0211	0.0230	0.957
	16000	0.0317	0.0097	0.0138	0.0152	0.923
$d_x = 50$ ($\gamma = 3$)	4000	0.0997	−0.0039	0.0377	0.0432	0.980
	8000	0.0702	0.0078	0.0202	0.0261	0.980
	16000	0.0484	0.0049	0.0152	0.0175	0.983
$d_x = 100$ ($\gamma = 5$)	4000	0.1221	0.0161	0.0392	0.0471	0.977
	8000	0.0905	0.0148	0.0225	0.0290	0.967
	16000	0.0631	0.0005	0.0154	0.0197	0.983
<i>Panel B: Value V (early-stopping boosting, γ_V)</i>						
$d_x = 10$ ($\gamma = 3$)	4000	0.0028	0.0017	0.0317	0.0325	0.943
	8000	0.0084	0.0028	0.0202	0.0202	0.937
	16000	0.0139	0.0049	0.0138	0.0137	0.927
$d_x = 50$ ($\gamma = 2$)	4000	0.0522	−0.0079	0.0391	0.0392	0.960
	8000	0.0404	−0.0075	0.0229	0.0232	0.947
	16000	0.0339	0.0021	0.0154	0.0152	0.943
$d_x = 100$ ($\gamma = 1.5$)	4000	0.0875	−0.0025	0.0409	0.0424	0.957
	8000	0.0731	−0.0040	0.0266	0.0253	0.940
	16000	0.0557	−0.0077	0.0169	0.0162	0.913

Notes: “Plug-in bias” is the average deviation of the uncorrected ML plug-in; the remaining columns refer to the cross-fitted sieve-debiased estimator $\tilde{\theta}$, its sampling SD, the mean studentized SE $\hat{\sigma}_n/\sqrt{n}$, and empirical 95% coverage over 300 iterations. Truths are d_x -stable: $W(h_0) \approx 0.34$, $V(h_0) \approx 0.585$. The sieve frequency rises for Welfare ($\gamma_W = 3, 3, 5$) and falls for Value ($\gamma_V = 3, 2, 1.5$) as $d_x = 10, 50, 100$, reflecting the regular/irregular divergence discussed in the text.

earnings, years of education, and treatment status. Our objective is to estimate and conduct inference on the first-best welfare functional and the optimal fraction of the population that should receive treatment.

As the first step in the estimation and inference procedure, we trim observations outside the common support of the treated and control groups to enforce the strict overlap assumption and to avoid extrapolation when applying sieve estimators. For either outcome measure, we then estimate $\mu_0(x, 1)$ nonparametrically using B-spline sieves fitted on the treated sample, and $\mu_0(x, 0)$ analogously on the control sample. The sieve dimensions are selected in a data-driven manner following [Chen, Christensen and Kankanala \(2025\)](#). Evaluating these estimates on the common support yields the CATE estimate $\hat{h}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$. Taking sample averages of $[\hat{h}(x)]_+$ and $\mathbb{1}\{\hat{h}(x) \geq 0\}$ over the trimmed dataset produces the estimators $\hat{W}(\hat{\mu})$ and $\hat{V}(\hat{\mu})$, respectively. Confidence intervals for $\bar{W}(\mu_0)$ and $\bar{V}(\mu_0)$ are then constructed according to their asymptotic theories: $\left(\hat{W}(\hat{\mu}) \pm 1.96 \hat{\sigma}_W / \sqrt{N}\right)$, and $\left(\hat{V}(\hat{\mu}) \pm 1.96 \hat{\sigma}_V / \sqrt{N}\right)$, where N denotes the sample size of the JTPA dataset.

While $\hat{\sigma}_W$ can be obtained directly using a plug-in estimate, the estimation of $\hat{\sigma}_V$ involves several additional steps. Specifically, it requires a density estimate for the covariates in the trimmed dataset. To this end, we use a Gaussian kernel density estimator, selecting the bandwidth matrix via the smoothed cross-validation method (`Hscv()` in the `ks` package) and applying a scaling factor of $s = 3$ to ensure adequate smoothness in the presence of discrete values for years of education. Furthermore, we need to specify a small hyperparameter ϵ to provide a close approximation to the pathwise derivative of the value functional in $\hat{\sigma}_V$. We set ϵ equal to a fraction $\iota = 0.01$ of the standard deviation of $\hat{h}(x)$ over the trimmed dataset. Robustness checks on the tuning parameters s and ι are reported in [Appendix D.2](#), and alternative density estimation approaches are examined in [Appendix D.3](#).

The results of our estimation and inference procedures are presented in [Table 10](#), alongside the nonparametric plug-in rule estimates from [Kitagawa and Tetenov \(2018\)](#), which serve as the most relevant comparison. Our estimates of both the welfare gain and the optimal treatment share are broadly consistent with theirs, despite methodological differences: we use sieve estimators for the nuisance parameters and trim the dataset to enforce common support, whereas they applied a Nadaraya-Watson estimator with an Epanechnikov kernel and did not trim. An important added contribution of our analysis is the provision of confidence intervals for both parameters of interest. By contrast, [Kitagawa and Tetenov \(2018\)](#) report confidence intervals only for the welfare gain under parametric rules and not for the optimal treatment share. Although our confidence interval for the welfare gain may appear wide, its length is comparable to those reported by [Kitagawa and Tetenov \(2018\)](#).

under parametric rules—underscoring that our inference framework remains sharp even while relying on nonparametric methods.

Table 10: Estimated Welfare Gains and Share of Population to be Treated Under Nonparametric Plug-in Rule

(a) 30-Month Post-Program Earnings, No Treatment Cost		
Method	Share Treated	Est. Welfare Gain
Ours	0.89 (0.73, 1.05)	\$1,519 (\$764, \$2,274)
KT (2018)	0.91 NA	\$1,693 NA
(b) 30-Month Post-Program Earnings, \$774 Cost per Treatment		
Method	Share Treated	Est. Welfare Gain
Ours	0.80 (0.53, 1.07)	\$858 (\$152, \$1,564)
KT (2018)	0.78 NA	\$996 NA

Note: Two-sided 95% confidence intervals in parentheses, constructed based on the asymptotic distributions of the corresponding estimators.

Two features of Table 10 warrant comment, and both reflect the distribution of the outcome variable rather than the inference procedure. First, the confidence intervals for the welfare gain are wide; second, the upper endpoints of several confidence intervals for the optimal treatment share exceed one — for example (0.73, 1.05) and (0.53, 1.07) — which is logically impossible for a share. We report these intervals *verbatim*, as the honest output of the asymptotic-normal construction, rather than truncating them. Both features trace to the 30-month earnings outcome, which is heavily right-skewed, with a mass of zeros and extreme upper outliers: its maximum is roughly ten times its mean, its standard deviation exceeds its mean, and about 11.8% of individuals report exactly zero earnings (see Figure 1 and Appendix D.1). We deliberately use the Kitagawa and Tetenov (2018) replication package without further processing the outcome, so that our results remain directly comparable to theirs; consequently the welfare-gain intervals inherit the large outcome variance and are wide, but comparable in length to those Kitagawa and Tetenov (2018) report for parametric rules in their Table 1. For the optimal treatment share, the functional is irregular (estimated

at a slower-than- $\sqrt{n}$ rate) and the point estimates lie close to one; together with the large standard error induced by the skewed outcome, the symmetric interval $\hat{V}(\hat{\mu}) \pm 1.96 \hat{\sigma}_V / \sqrt{N}$ then extends beyond one. This is a property of the data and of the untransformed normal approximation near the boundary of the parameter space, not of the point estimator: a reader may simply intersect the reported interval with $[0, 1]$ without altering the point estimate.

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A Main Proofs

Proof of Theorem 1. For any ν s.t. $\int \nu^2(x) f(x) dx < \infty$, write $h_t := h_0 + t\nu$ and consider the functional derivative

$$\begin{aligned} D_h \overline{W}(h_0)[\nu] &= \left. \frac{d}{dt} \overline{W}(h_t) \right|_{t=0} \\ &= \left. \frac{d}{dt} \int [h_t(x)]_+ f(x) dx \right|_{t=0}. \end{aligned}$$

We first show the interchangeability of the differentiation and integration holds:

$$\left. \frac{d}{dt} \int [h_t(x)]_+ f(x) dx \right|_{t=0} = \int \left. \frac{\partial}{\partial t} [h_t(x)]_+ \right|_{t=0} f(x) dx.$$

Notice that $[h_t(x)]_+$ is Lipschitz in t :

$$|[h_t(x)]_+ - [h_s(x)]_+| \leq |\nu(x)| |t - s|$$

with

$$\int |\nu(x)| f(x) dx < \infty.$$

Furthermore, $[h_t(x)]_+$ is almost everywhere differentiable in t with

$$\left. \frac{\partial}{\partial t} [h_t(x)]_+ \right|_{t=0} = \mathbb{1}\{h_0(x) \geq 0\} \nu(x) \quad (27)$$

for any x s.t. $h_0(x) \neq 0$. Since $\{x : h_0(x) = 0\}$ has Lebesgue measure 0 under Assumption 2(c), we have

$$\mathbb{P}_f(h_0(X_i) = 0) = 0.$$

Hence, (27) holds a.s.- f in x , and thus by the dominated convergence theorem, we have

$$\begin{aligned} \left. \frac{d}{dt} \int [h_t(x)]_+ f(x) dx \right|_{t=0} &= \int \left. \frac{\partial}{\partial t} [h_t(x)]_+ \right|_{t=0} f(x) dx \\ &= \int \mathbb{1}\{h_0(x) \geq 0\} \nu(x) f(x) dx. \end{aligned} \quad (28)$$

We now switch the notation from h_0 to μ_0 for subsequent analysis, and consider the functional derivative of $\overline{W}(\mu_0)$ w.r.t. μ in the direction of ν . Note that $\mu(x, d)$ and $\nu(x, d)$ are functions of both x and d . Applying (28), we have

$$\begin{aligned} D_\mu \overline{W}(\mu_0)[\nu] &:= \left. \frac{d}{dt} \overline{W}(\mu_0 + t\nu) \right|_{t=0} \\ &= \int \mathbb{1}\{\mu_0(x, 1) - \mu_0(x, 0) \geq 0\} (\nu(x, 1) - \nu(x, 0)) f(x) dx \\ &= \int \mathbb{1}\{h_0(x) \geq 0\} (\nu(x, 1) - \nu(x, 0)) \lambda(x) f_0(x) dx \\ &= \mathbb{E}[\mathbb{1}\{h_0(X_i) \geq 0\} \lambda(X_i) (\nu(X_i, 1) - \nu(X_i, 0))] \\ &= \mathbb{E}\left[\mathbb{1}\{h_0(X_i) \geq 0\} \lambda(X_i) \left(\frac{D_i}{p_0(X_i)} \nu(X_i, D_i) - \frac{1 - D_i}{1 - p_0(X_i)} \nu(X_i, D_i)\right)\right] \\ &= \mathbb{E}\left[\mathbb{1}\{h_0(X_i) \geq 0\} \lambda(X_i) \left(\frac{D_i}{p_0(X_i)} - \frac{1 - D_i}{1 - p_0(X_i)}\right) \nu(X_i, D_i)\right] \\ &= \mathbb{E}[\nu^*(X_i, D_i) \nu(X_i, D_i)] \end{aligned} \quad (29)$$

where $\lambda := f/f_0$, $\mathbb{E}[\cdot]$ is expectation taken with respect to the training data distribution, and

$$\nu^*(x, d) := \mathbb{1}\{h_0(x) \geq 0\} \lambda(x) \left(\frac{d}{p_0(x)} - \frac{1 - d}{1 - p_0(x)}\right) \quad (30)$$

is the Riesz representer for the linear functional $D_\mu \overline{W}(\mu_0)[\cdot]$. Since

$$\begin{aligned} &\mathbb{E}\left[\left(\frac{D_i}{p_0(X_i)} - \frac{1 - D_i}{1 - p_0(X_i)}\right)^2 \middle| X_i\right] \\ &= \mathbb{E}\left[\left(\frac{D_i - D_i p_0(X_i) - p_0(X_i) + D_i p_0(X_i)}{p_0(X_i)(1 - p_0(X_i))}\right)^2 \middle| X_i\right] \end{aligned}$$

$$\begin{aligned}
&= \frac{\mathbb{E} \left[(D_i - p_0(X_i))^2 \middle| X_i \right]}{p_0^2(X_i) (1 - p_0(X_i))^2} = \frac{p_0(X_i) (1 - p_0(X_i))}{p_0^2(X_i) (1 - p_0(X_i))^2} \\
&= \frac{1}{p_0(X_i) (1 - p_0(X_i))}
\end{aligned}$$

under the overlap condition in Assumption 1(b), the Riesz representer ν^* has finite norm

$$\begin{aligned}
\|\nu^*\|^2 &= \mathbb{E} \left[\mathbb{1} \{h_0(X_i) \geq 0\} \lambda^2(X_i) \left(\frac{D_i}{p_0(X_i)} - \frac{1 - D_i}{1 - p_0(X_i)} \right)^2 \right] \\
&= \mathbb{E} \left[\frac{\mathbb{1} \{h_0(X_i) \geq 0\} \lambda^2(X_i)}{p_0(X_i) (1 - p_0(X_i))} \right] < \infty
\end{aligned}$$

showing that $D_\mu \overline{W}(\mu_0)$ is a regular linear functional.

Next, we control the remainder term from the linearization. Specifically, note that

$$D_h W(h) [\nu] = \int \mathbb{1} \{h(x) \geq 0\} \nu(x) f(x) dx$$

which is exactly of the form of the generic value functional. We then have

$$\begin{aligned}
D_h^2 W(h_0) [\nu, u] &= \int_{\{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}} \frac{1}{\|\nabla_x h_0(x)\|} \nu(x) u(x) f(x) d\mathcal{H}^{d_x-1}(x) \\
&\leq \frac{1}{\underline{c}} \|\nu\|_\infty \|u\|_\infty \int_{\{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}} f(x) d\mathcal{H}^{d_x-1}(x) =: C_0 \|\nu\|_\infty \|u\|_\infty,
\end{aligned}$$

where $C_0 < \infty$ under Assumption 2(c) and the boundedness of f (via the co-area formula).

Similar bound applies to $D_\mu^2 \overline{W}(h_0) [\nu, u]$ as well. Thus we obtain:

$$\left| \overline{W}(\hat{\mu}) - \overline{W}(\mu_0) - D_\mu \overline{W}(\mu_0) [\hat{\mu} - \mu_0] \right| \leq M \|\hat{\mu} - \mu_0\|_\infty^2.$$

Next, by Assumption 3(b), the linear term admits the asymptotic linear representation

$$\sqrt{n} D_\mu \overline{W}(\mu_0) [\hat{\mu} - \mu_0] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \nu^*(X_i, D_i) \epsilon_i + o_p(1),$$

which holds, in particular, for the linear sieve least squares first stage of Lemma 1 (with the boundedness of ν^* just established and undersmoothing controlling the sieve bias). Hence, we have

$$\begin{aligned}
\sqrt{n} \left(\overline{W}(\hat{\mu}) - \overline{W}(\mu_0) \right) &= \sqrt{n} D_\mu \overline{W}(\mu_0) [\hat{\mu} - \mu_0] + \sqrt{n} O_p \left(\|\hat{\mu} - \mu_0\|_\infty^2 \right) \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^n \nu^*(X_i, D_i) \epsilon_i + o_p(1) + \sqrt{n} o_p \left(n^{-1/2} \right) \\
&\xrightarrow{d} \mathcal{N} \left(0, \sigma_W^2 \right)
\end{aligned}$$

where, writing $\sigma_\epsilon^2(x, d) := \mathbb{E}[\epsilon_i^2 | X_i = x, D_i = d]$ and noting that $D_i(1 - D_i) = 0$,

$$\sigma_W^2 := \text{Var}(\nu^*(X_i, D_i) \epsilon_i) = \mathbb{E} \left[\mathbb{1} \{h_0(X_i) \geq 0\} \lambda^2(X_i) \left(\frac{D_i}{p_0^2(X_i)} + \frac{1 - D_i}{(1 - p_0(X_i))^2} \right) \epsilon_i^2 \right]$$

$$\begin{aligned}
&= \mathbb{E} \left[\mathbb{1} \{h_0(X_i) \geq 0\} \lambda^2(X_i) \left(\frac{\sigma_\epsilon^2(X_i, 1)}{p_0(X_i)} + \frac{\sigma_\epsilon^2(X_i, 0)}{1 - p_0(X_i)} \right) \right] \\
&= \int \mathbb{1} \{h_0(x) \geq 0\} \lambda(x) \left(\frac{\sigma_\epsilon^2(x, 1)}{p_0(x)} + \frac{\sigma_\epsilon^2(x, 0)}{1 - p_0(x)} \right) f(x) dx.
\end{aligned}$$

□

Proof of Theorem 2. We apply the derivations in Theorem 1 with $f = f_0$ and consequently $\lambda \equiv 1$.

Consider the following standard empirical process decomposition

$$\sqrt{n}(\mathbb{P}_n g_{\hat{\mu}} - P g_{\mu_0}) = \sqrt{n}P(g_{\hat{\mu}} - g_{\mu_0}) + \mathbb{G}_n g_{\mu_0} + \mathbb{G}_n(g_{\hat{\mu}} - g_{\mu_0}), \quad (31)$$

where $g_\mu(x) := [\mu(x, 1) - \mu(x, 0)]_+$, $\mathbb{P}_n g := \frac{1}{n} \sum_{i=1}^n g$, $Pg = \int g dF_0$ and $\mathbb{G}_n := \sqrt{n}(\mathbb{P}_n - P)$. Note that the term $\sqrt{n}P(g_{\hat{\mu}} - g_{\mu_0})$ corresponds to the analysis of the population expectation (integral) with respect to the true distribution $F = F_0$ in the last subsection. There are two additional terms that appear due to the use of the sample average: a “stochastic equicontinuity” term $\mathbb{G}_n(g_{\hat{\mu}} - g_{\mu_0})$ that will be shown to be asymptotically negligible under the permanence of the Donsker property, as well as a “CLT”-term $\mathbb{G}_n g_{\mu_0}$ that adds to the asymptotic variance of the estimator.

Under Assumption 3(c), $\hat{\mu} - \mu_0$ belongs to a Donsker class of functions, and the Lipschitz transformation of μ_0 through the ReLU function $[\cdot]_+$ preserves the Donsker property. Hence, $g_{\hat{\mu}} - g_{\mu_0}$ also belongs to a Donsker class; together with $\|g_{\hat{\mu}} - g_{\mu_0}\|_{L^2(P)} \leq 2\|\hat{\mu} - \mu_0\|_\infty = o_p(1)$ (implied by Assumption 3(a)), the asymptotic equicontinuity of the empirical process yields $\mathbb{G}_n(g_{\hat{\mu}} - g_{\mu_0}) = o_p(1)$.

Then, by (31) we have,

$$\begin{aligned}
&\sqrt{n}(\hat{W}(\hat{h}) - W(h_0)) \equiv \sqrt{n}(\hat{\bar{W}}(\hat{\mu}) - \bar{W}(\mu_0)) \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^n ([h_0(X_i)]_+ - W(h_0) + \nu^*(X_i, D_i) \epsilon_i) + o_p(1) \xrightarrow{d} \mathcal{N}(0, \bar{\sigma}_W^2)
\end{aligned}$$

where

$$\begin{aligned}
\bar{\sigma}_W^2 &:= \text{Var}([h_0(X_i)]_+ - W(h_0) + \nu^*(X_i, D_i) \epsilon_i) \\
&= \text{Var}([h_0(X_i)]_+) + \text{Var}(\nu^*(X_i, D_i) \epsilon_i) + 2\text{Cov}([h_0(X_i)]_+, \nu^*(X_i, D_i) \epsilon_i) \\
&= \text{Var}([h_0(X_i)]_+) + \text{Var}(\nu^*(X_i, D_i) \epsilon_i) \\
&= \text{Var}([h_0(X_i)]_+) + \sigma_W^2.
\end{aligned}$$

□

Proof of the closed-form (AIPW) construction of Remark 4. We treat the unknown-density

case (the AIPW form of m in (25)); the known-density case is analogous and simpler, with the term $\text{Var}([h_0]_+)$ absent. Fix a fold κ , write $\hat{\eta} := \hat{\eta}_{-\kappa}$ and $\mathbb{E}_{n,\kappa}[g] := \frac{\kappa}{n} \sum_{i \in I_\kappa} g$, and condition on the data outside fold κ , so that $\hat{\eta}$ is fixed and independent of $\{Z_i\}_{i \in I_\kappa}$. Decompose

$$\mathbb{E}_{n,\kappa}[\psi(\cdot; \theta_0, \hat{\eta})] = \mathbb{E}_{n,\kappa}[\psi(\cdot; \theta_0, \eta_0)] + \underbrace{(\mathbb{E}_{n,\kappa} - \mathbb{E})[\psi(\cdot; \theta_0, \hat{\eta}) - \psi(\cdot; \theta_0, \eta_0)]}_{(a)} + \underbrace{\mathbb{E}[\psi(\cdot; \theta_0, \hat{\eta})]}_{(b)},$$

where the outer $\mathbb{E}$ in (b) integrates over an independent copy of Z holding $\hat{\eta}$ fixed. By cross-fitting and the consistency (ii), $\|\psi(\cdot; \theta_0, \hat{\eta}) - \psi(\cdot; \theta_0, \eta_0)\|_{L^2(P)} = o_p(1)$, so the conditional Chebyshev inequality gives (a) $= o_p(n^{-1/2})$. For (b), moment validity gives $\mathbb{E}[\psi(\cdot; \theta_0, \eta_0)] = 0$, and Neyman orthogonality kills the first-order term in the expansion of $\hat{\eta} \mapsto \mathbb{E}[\psi(\cdot; \theta_0, \hat{\eta})]$ about η_0 , leaving a second-order remainder

$$|(b)| \leq C \left(\|\hat{\mu} - \mu_0\|_2 \left(\|\hat{p} - p_0\|_2 + \|\hat{\lambda} - \lambda\|_2 \right) + \|\hat{\mu} - \mu_0\|_\infty^2 \right) = o_p(n^{-1/2})$$

under the product rate (iii). In detail, writing $\delta_d := \hat{\mu}(\cdot, d) - \mu_0(\cdot, d)$, the orthogonality computation in the unknown-density case yields (b) $= T_1 + T_2 + T_3$ with

$$\begin{aligned} T_1 &= \mathbb{E} \left[\left(\mathbb{1}\{h_0 \geq 0\} - \mathbb{1}\{\hat{h} \geq 0\} \right) (\delta_1 - \delta_0) \right], \\ T_2 &= \mathbb{E} \left[\mathbb{1}\{\hat{h} \geq 0\} (\hat{p} - p_0) \left(\frac{\delta_1}{\hat{p}} + \frac{\delta_0}{1 - \hat{p}} \right) \right], \\ T_3 &= \mathbb{E} \left[[\hat{h}]_+ - [h_0]_+ - \mathbb{1}\{h_0 \geq 0\} (\hat{h} - h_0) \right] \end{aligned}$$

(the density ratio contributing an analogous bilinear term in the known-density case). The boundary term T_1 is genuinely second order: a sign flip between $\hat{h}(x)$ and $h_0(x)$ forces $|h_0(x)| \leq 2\|\hat{\mu} - \mu_0\|_\infty$ and simultaneously $|\delta_1(x) - \delta_0(x)| \leq 2\|\hat{\mu} - \mu_0\|_\infty$, so $|T_1| \leq 2\|\hat{\mu} - \mu_0\|_\infty \mathbb{P}(|h_0(X)| \leq 2\|\hat{\mu} - \mu_0\|_\infty) \leq M\|\hat{\mu} - \mu_0\|_\infty^2$ by the boundedness of f_0 (co-area). Crucially, T_1 carries *no* first-order propensity error: the propensity enters only through the bilinear $T_2 = O(\|\hat{p} - p_0\|_2 \|\hat{\mu} - \mu_0\|_2)$, by Cauchy–Schwarz under the strong overlap of Assumption 1(b). Finally $T_3 = O(\|\hat{\mu} - \mu_0\|_\infty^2)$ is the ReLU curvature remainder, controlled exactly as in the bound on $D_h^2 W$ in the proof of Theorem 1. Since $\hat{\theta}_{DML}$ solves $\frac{1}{\kappa} \sum_{\kappa} \mathbb{E}_{n,\kappa}[\psi(\cdot; \hat{\theta}_{DML}, \hat{\eta}_{-\kappa})] = 0$ and ψ is linear in θ with coefficient -1 , averaging the above over the κ folds yields

$$\hat{\theta}_{DML} - \theta_0 = \frac{1}{n} \sum_{i=1}^n \psi(Z_i; \theta_0, \eta_0) + o_p(n^{-1/2}),$$

where $\psi(Z_i; \theta_0, \eta_0) = [h_0(X_i)]_+ - W(h_0) + \nu^*(X_i, D_i) \epsilon_i$ has mean zero and variance $\bar{\sigma}_W^2$ (Theorem 2), the cross term vanishing because $\mathbb{E}[\epsilon_i | X_i, D_i] = 0$. The Lindeberg–Feller central limit theorem delivers $\sqrt{n}(\hat{\theta}_{DML} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \bar{\sigma}_W^2)$, and $\hat{\sigma}^2 \xrightarrow{p} \bar{\sigma}_W^2$ by the law of large numbers for the cross-fitted scores together with the consistency in (ii). In the

known-density case the same argument applies with $\psi(Z_i; \theta_0, \eta_0) = \nu^*(X_i, D_i) \epsilon_i$ and limiting variance σ_W^2 . $\square$

Proof of Theorem 3. Recall that

$$\theta_0 = V(h_0) := \int \mathbb{1}\{h_0(x) \geq 0\} v_0(x) f(x) dx, \quad \hat{\theta} = V(\hat{h}) = \bar{V}(\hat{\mu}).$$

Taking the functional derivative according to [Chen and Gao \(2026\)](#), we obtain the following submanifold integral with submanifold dimension $m = d_x - 1$,

$$D_h V(h_0)[\nu] := \int_{\{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}} \frac{\nu(x)}{\|\nabla_x h_0(x)\|} v_0(x) f(x) d\mathcal{H}^{d_x-1}(x).$$

Equivalently, using the notation μ_0 , we have

$$D_\mu \bar{V}(\mu_0)[\nu] := \int_{\{x \in \mathbb{R}^{d_x} : h_0(x) = 0\}} \frac{(\nu(x, 1) - \nu(x, 0))}{\|\nabla_x h_0(x)\|} v_0(x) f(x) d\mathcal{H}^{d_x-1}(x).$$

Applying Lemma 3 (pathwise derivative of upper-contour integrals), Theorem 5, and Propositions 1–2 in [Chen and Gao \(2026\)](#), we obtain that

$$\frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\sigma_{V,n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{with } \sigma_{V,n}^2 \asymp K_n^{\frac{1}{d_x}}.$$

$\square$

Proof of Theorem 4. We now work with the following rescaled empirical process decomposition

$$\sqrt{\frac{n}{\sigma_{V,n}^2}} (\mathbb{P}_n g_{\hat{\mu}} - P g_{\mu_0}) = \sqrt{\frac{n}{\sigma_{V,n}^2}} P(g_{\hat{\mu}} - g_{\mu_0}) + \sigma_{V,n}^{-1} \mathbb{G}_n g_{\mu_0} + \sigma_{V,n}^{-1} \mathbb{G}_n (g_{\hat{\mu}} - g_{\mu_0}), \quad (32)$$

where $g_\mu(x) := \mathbb{1}\{\mu(x, 1) - \mu(x, 0) \geq 0\} v_0(x)$.

Note that the term $P(g_{\hat{\mu}} - g_{\mu_0}) \equiv \bar{V}(\hat{\mu}) - \bar{V}(\mu_0)$ has been analyzed in Section 4.1, where we have shown that

$$\sqrt{\frac{n}{\sigma_{V,n}^2}} P(g_{\hat{\mu}} - g_{\mu_0}) \xrightarrow{d} \mathcal{N}(0, 1).$$

Given the above, the term $\sigma_{V,n}^{-1} \mathbb{G}_n g_{\mu_0} = O_p\left(\sqrt{K_n^{-1/d_x}}\right) = o_p(1)$ becomes asymptotically negligible. Below we seek to show that, in fact, the last term is also asymptotically negligible:

$$\sigma_{V,n}^{-1} \mathbb{G}_n (g_{\hat{\mu}} - g_{\mu_0}) = o_p(1).$$

Note that $g_\mu(x)$ involves a discontinuous indicator function, which does not preserve the Donsker property for the Hölder class in general.⁶ We thus directly derive the required

⁶In contrast, the indicator function transformation preserves the Donsker property for VC classes of functions, in which case the Donsker property would deliver $\mathbb{G}_n(g_{\hat{\mu}} - g_{\mu_0}) = o_p(1)$, which implies the weaker condition $K_n^{-1/d_x} \mathbb{G}_n(g_{\hat{\mu}} - g_{\mu_0}) = o_p(1)$ required in this paper.

negligibility via a maximal inequality on the functional class

$$\mathcal{G}_{a_n} := \{g_\mu - g_{\mu_0} : \|\mu - \mu_0\|_\infty \leq a_n\},$$

where $a_n \searrow 0$ is a deterministic sequence with $\|\hat{\mu} - \mu_0\|_\infty = O_p(a_n)$ (such a sequence exists since $\|\hat{\mu} - \mu_0\|_\infty = o_p(1)$ under Assumption 4).

We first obtain an envelope function for $\mathcal{G}_{a_n}$ and its second moment:

$$\begin{aligned} |g_\mu(x) - g_{\mu_0}(x)| &= |v_0(x)| |\mathbb{1}\{\mu(x, 1) - \mu(x, 0) \geq 0\} - \mathbb{1}\{\mu_0(x, 1) - \mu_0(x, 0) \geq 0\}| \\ &\leq \|v_0\|_\infty \left(\mathbb{1}\{\mu(x, 1) - \mu(x, 0) \geq 0 > \mu_0(x, 1) - \mu_0(x, 0)\} \right. \\ &\quad \left. + \mathbb{1}\{\mu_0(x, 1) - \mu_0(x, 0) \geq 0 > \mu(x, 1) - \mu(x, 0)\} \right) \\ &\leq \|v_0\|_\infty \mathbb{1}\{|\mu_0(x, 1) - \mu_0(x, 0)| \leq 2\|\mu - \mu_0\|_\infty\} \\ &\leq \|v_0\|_\infty \mathbb{1}\{|\mu_0(x, 1) - \mu_0(x, 0)| \leq 2a_n\} =: G_{a_n}, \end{aligned}$$

where the third inequality uses that a sign change between $\mu(x, 1) - \mu(x, 0)$ and $\mu_0(x, 1) - \mu_0(x, 0)$ forces $|h_0(x)| \leq 2\|\mu - \mu_0\|_\infty$. Its second moment satisfies

$$PG_{a_n}^2 = \|v_0\|_\infty^2 \mathbb{P}(|h_0(X)| \leq 2a_n) \leq Ma_n,$$

where the last inequality holds under Assumption 2(c), the boundedness of v_0 , and the boundedness of f_0 (via the co-area formula). It remains to bound the uniform entropy of $\mathcal{G}_{a_n}$. Restricted to $\|\mu - \mu_0\|_\infty \leq a_n$ with a_n small, the upper-contour sets $\{x : \mu(x, 1) - \mu(x, 0) \geq 0\}$ have boundaries that are uniformly small C^1 perturbations of the regular level set $\{x : h_0(x) = 0\}$: since $\|\nabla_x h_0\| \geq \underline{c} > 0$ there (Assumption 2(c)) and $h_0 \in \Lambda^s$, by the implicit function theorem each such boundary is, locally, the graph of an s -Hölder function, uniformly over the neighborhood. By the classical metric-entropy bound for indicators of sets with s -Hölder boundaries (the *boundary-fragment* classes of ? and ?; see also Section 2.7.2 of Van Der Vaart and Wellner, 1996), the L^2 covering numbers satisfy $\log N(\epsilon) \lesssim \epsilon^{-(d_x-1)/s}$, so the uniform entropy integral J_G of $\mathcal{G}_{a_n}$ is finite whenever $(d_x - 1)/s < 2$, i.e. $s > (d_x - 1)/2$ — which is implied by the maintained smoothness $s > d_x$ of Theorem 3. Hence, with this finite J_G ,

$$P \sup_{\|\mu - \mu_0\|_\infty \leq a_n} |\mathbb{G}_n(g_\mu - g_{\mu_0})| \leq J_G \sqrt{PG_{a_n}^2} \leq M\sqrt{a_n}$$

and thus

$$\sigma_{V,n}^{-1} \mathbb{G}_n(g_{\hat{\mu}} - g_{\mu_0}) = O_p\left(\sqrt{K_n^{-1/d_x} a_n}\right) = o_p(1).$$

Hence,

$$\begin{aligned}\sqrt{\frac{n}{\sigma_{V,n}^2}} \left(\hat{V}(\hat{h}) - V(h_0) \right) &\equiv \sqrt{\frac{n}{\sigma_{V,n}^2}} \left(\hat{\bar{V}}(\hat{\mu}) - \bar{V}(\mu_0) \right) \\ &= \sqrt{\frac{n}{\sigma_{V,n}^2}} \left(\bar{V}(\hat{\mu}) - \bar{V}(\mu_0) \right) + o_p(1) \\ &\xrightarrow{d} \mathcal{N}(0, 1).\end{aligned}$$

□

Proof of Theorem 5. It suffices to establish the lower bound over a subclass of data generating processes permitted by the hypotheses of the theorem. Consider the subclass $\mathcal{P}_0$ of DGPs in which: (i) $\mu_0(\cdot, 0) \equiv 0$; (ii) $p_0(x) = 1/2$ for all $x \in \mathcal{X}$; (iii) $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent of (X_i, D_i) for some fixed $\sigma^2 > 0$; and (iv) $\mu_0(\cdot, 1) = h_0 \in \Lambda_c^s(\mathcal{X})$. For any DGP in $\mathcal{P}_0$, the value functional becomes

$$V(h_0) = \int \mathbb{1}\{h_0(x) \geq 0\} w(x) dx, \quad w := v_0 \cdot f,$$

which is exactly the upper-contour integral functional analyzed in Lemma 3 and Corollary 2 of [Chen and Gao \(2026\)](#), with their weight regularity and linearization conditions satisfied under hypotheses (i)–(iii) of the present theorem together with Assumption 2(c).

For any two $h, h' \in \Lambda_c^s(\mathcal{X})$ indexing DGPs in $\mathcal{P}_0$, the Kullback–Leibler divergence between the corresponding joint distributions of the training data factorizes as

$$KL(P_h^n \| P_{h'}^n) = n \mathbb{E} \left[\frac{p_0(X_i) (h(X_i) - h'(X_i))^2}{2\sigma^2} \right] = \frac{n}{2} \mathbb{E} \left[\frac{(h(X_i) - h'(X_i))^2}{2\sigma^2} \right],$$

since the untreated observations (with $D_i = 0$) are uninformative about $h_0 = \mu_0(\cdot, 1)$ within $\mathcal{P}_0$. Hence the per-observation KL divergence in the treatment model equals exactly one half of that in the nonparametric regression model $Y_i = h_0(X_i) + \epsilon_i$ used in the lower bound construction of [Chen and Gao \(2026\)](#). Since their Le Cam-type lower bound argument depends on the DGPs only through such KL divergences, the same construction applies to the treatment model with the constant adjusted by a factor of 1/2, leaving the rate $r_n^* = n^{-\frac{s}{2s+1}}$ unchanged. Finally, as $\mathcal{P}_0$ is contained in the class of DGPs permitted by the hypotheses of the theorem, the minimax risk over the full class is bounded below by that over $\mathcal{P}_0$, which completes the proof. □

Proof of Theorem 6. The result follows by applying Theorem 5 and Propositions 1–2 of [Chen and Gao \(2026\)](#) for the SS and LOO estimators of upper-contour integrals, along the same lines as the proof of Theorem 3. The only adaptation required concerns the difference-of-regressions structure of the CATE estimator: in the stacked sieve regression

(17), the regressor vector $\bar{\psi}(X_i, D_i)$ plays the role of the sieve basis, and $\hat{h}(x) = a(x)' \hat{\beta}$ with $a(x) = (\psi^{(K_1)}(x)', -\psi^{(K_0)}(x)')'$. Consequently, $\hat{s}_i(x) = a(x)' \hat{G}^{-1} \bar{\psi}(X_i, D_i)$ is the empirical influence of observation i on $\hat{h}(x)$, and the leave-one-out residuals are computed from the stacked regression, so that the diagonal-part corrections in (18) and (19) coincide with those in Chen and Gao (2026) applied to the stacked regression. $\square$

B DML via the Sieve Riesz Representer: Construction and Proofs

Section 5 introduced, for a generic functional Θ , the cross-fitted sieve-debiased estimator $\tilde{\theta}$ in (24), built from the sieve Riesz representer $\hat{v}_{K_n}^*$ in (23). This appendix records the local-robustness conditions (Assumption 5), proves the general asymptotic normality result (Theorem 7), and specializes it to the welfare and value functionals (Corollaries 1 and 2).

We work with the treatment assignment model of Section 2.1, a generic functional Θ with $\theta_0 := \Theta(\mu_0)$, and the Hilbert space $\mathcal{V}$ of Section 5.1, with inner product $\langle \cdot, \cdot \rangle$, induced norm $\|\cdot\|$, standard-deviation norm $\|\cdot\|_{sd}$, and $\sigma_\epsilon^2(x, d) := \mathbb{E}[\epsilon_i^2 | X_i = x, D_i = d]$. Recall the augmented functional $\tilde{\Theta}(\mu, v)$ of (20), whose derivative $D_\mu \tilde{\Theta}(\mu_0, v)[\nu] = D_\mu \Theta(\mu_0)[\nu] - \langle v, \nu \rangle$ is given in (21). On the finite-dimensional sieve space $\mathcal{V}_{K_n}$ spanned by the components of the stacked regressor $\bar{\psi}$ of (17), with $\langle \cdot, \cdot \rangle$ -orthogonal projection Π_{K_n} , the sieve Riesz representer $v_{K_n}^*$ of (22) satisfies $D_\mu \Theta(\mu_0)[\nu] = \langle v_{K_n}^*, \nu \rangle$ for all $\nu \in \mathcal{V}_{K_n}$, hence $D_\mu \tilde{\Theta}(\mu_0, v_{K_n}^*)[\nu] \equiv 0$ on $\mathcal{V}_{K_n}$: the first-order bias is annihilated locally on the sieve space, whether or not a global Riesz representer exists. Write $\sigma_n := \|v_{K_n}^*\|_{sd}$. We impose the following conditions on the estimation errors of μ_0 and $v_{K_n}^*$.

Assumption 5 (Local Robustness). *For each $\kappa = 1, \dots, K$:*

- (a) *Errors:* $0 < \inf_{x,d} \sigma_\epsilon^2(x, d) \leq \sup_{x,d} \sigma_\epsilon^2(x, d) < \infty$, and $\sup_{x,d} \mathbb{E}[\epsilon_i^2 \mathbb{1}\{|\epsilon_i| > c\} | X_i = x, D_i = d] \rightarrow 0$ as $c \rightarrow \infty$.
- (b) *Consistency:* $\|\hat{\mu}_{-\kappa} - \mu_0\|_\infty = o_p(1)$ and $\|\hat{v}_{K_n, -\kappa}^* - v_{K_n}^*\| = o_p(\sigma_n)$.
- (c) *Product Rate:* $\|\hat{v}_{K_n, -\kappa}^* - v_{K_n}^*\| \cdot \|\hat{\mu}_{-\kappa} - \mu_0\| = o_p(\sigma_n/\sqrt{n})$.
- (d) *Sieve Approximation Bias:* $|D_\mu \Theta(\mu_0)[(I - \Pi_{K_n})(\hat{\mu}_{-\kappa} - \mu_0)]| = o_p(\sigma_n/\sqrt{n})$.
- (e) *Linearization Remainder:* $|\Theta(\hat{\mu}_{-\kappa}) - \Theta(\mu_0) - D_\mu \Theta(\mu_0)[\hat{\mu}_{-\kappa} - \mu_0]| = o_p(\sigma_n/\sqrt{n})$.

Conditions (b) and (c) concern the estimation of the sieve Riesz representer and the first stage; they replace the requirement that $\hat{\mu}$ be the sieve least squares estimator, so that

$\hat{\mu}_{-\kappa}$ may be a generic machine-learning learner. Condition (d) controls how well the sieve space tracks the first-stage estimation error: it reduces to the standard (deterministic) sieve approximation bias $|D_\mu \Theta(\mu_0)[(I - \Pi_{K_n})\mu_0]|$ when $\hat{\mu}_{-\kappa} \in \mathcal{V}_{K_n}$, which is in turn controlled by undersmoothing the sieve dimension as in [Chen and Gao \(2026\)](#). Condition (e) is the linearization remainder requirement; for the value functional, Assumption 4 provides lower-level sufficient conditions. The common benchmark rate $\sigma_n/\sqrt{n}$ specializes to $1/\sqrt{n}$ when σ_n stays bounded (regular case) and to $\sqrt{K_n^{1/d_x}/n}$ when $\sigma_n \asymp K_n^{1/2d_x}$ (irregular case).

Proof of Theorem 7. Fix a fold κ and write $\hat{\mu} := \hat{\mu}_{-\kappa}$, $\hat{v}^* := \hat{v}_{K_n, -\kappa}^*$, $v^* := v_{K_n}^*$, and $\mathbb{E}_{n, \kappa}[g] := \frac{\kappa}{n} \sum_{i \in I_\kappa} g(Y_i, D_i, X_i)$. Since $\hat{\mu}$ and $\hat{v}^*$ are computed on the complement of fold κ , they are independent of the observations in I_κ , and below we condition on the data outside fold κ so that $\hat{\mu}$ and $\hat{v}^*$ may be treated as fixed. Writing $\tilde{\theta}_\kappa$ for the κ -th summand in (24), we decompose

$$\begin{aligned} \tilde{\theta}_\kappa - \theta_0 &= \underbrace{\Theta(\hat{\mu}) - \Theta(\mu_0) - D_\mu \Theta(\mu_0)[\hat{\mu} - \mu_0]}_{(R_1)} + D_\mu \Theta(\mu_0)[\hat{\mu} - \mu_0] \\ &\quad + \mathbb{E}_{n, \kappa}[v^* \epsilon] + \underbrace{\mathbb{E}_{n, \kappa}[(\hat{v}^* - v^*) \epsilon]}_{(R_2)} - \mathbb{E}_{n, \kappa}[\hat{v}^*(\hat{\mu} - \mu_0)], \end{aligned}$$

and further decompose the last term as

$$\mathbb{E}_{n, \kappa}[\hat{v}^*(\hat{\mu} - \mu_0)] = \langle v^*, \hat{\mu} - \mu_0 \rangle + \underbrace{\langle \hat{v}^* - v^*, \hat{\mu} - \mu_0 \rangle}_{(R_3)} + \underbrace{(\mathbb{E}_{n, \kappa} - \mathbb{E})[\hat{v}^*(\hat{\mu} - \mu_0)]}_{(R_4)}.$$

Moreover, since $v^* \in \mathcal{V}_{K_n}$ and the projection Π_{K_n} is self-adjoint,

$$D_\mu \Theta(\mu_0)[\hat{\mu} - \mu_0] - \langle v^*, \hat{\mu} - \mu_0 \rangle = D_\mu \Theta(\mu_0)[(I - \Pi_{K_n})(\hat{\mu} - \mu_0)] =: (R_5).$$

Collecting terms, $\tilde{\theta}_\kappa - \theta_0 = \mathbb{E}_{n, \kappa}[v^* \epsilon] + R_1 + R_2 - R_3 - R_4 + R_5$.

We now show $R_j = o_p(\sigma_n/\sqrt{n})$ for $j = 1, \dots, 5$. Terms R_1 and R_5 are $o_p(\sigma_n/\sqrt{n})$ by Assumption 5(e) and (d), respectively. By the Cauchy-Schwarz inequality and Assumption 5(c), $|R_3| \leq \|\hat{v}^* - v^*\| \|\hat{\mu} - \mu_0\| = o_p(\sigma_n/\sqrt{n})$. Conditionally on the data outside fold κ , R_2 has mean zero and variance at most $\frac{\kappa}{n} \left(\sup_{x, d} \sigma_\epsilon^2(x, d) \right) \|\hat{v}^* - v^*\|^2 = o_p(\sigma_n^2/n)$ by Assumption 5(a)(b), so $R_2 = o_p(\sigma_n/\sqrt{n})$ by the conditional Chebyshev inequality. Similarly, R_4 has conditional mean zero and conditional variance at most $\frac{\kappa}{n} \|\hat{\mu} - \mu_0\|_\infty^2 \|\hat{v}^*\|^2$; since $\|\hat{v}^*\| \leq \|v^*\| + \|\hat{v}^* - v^*\| = O_p(\sigma_n)$, where we use $\|v^*\| \asymp \|v^*\|_{sd} = \sigma_n$ under Assumption 5(a), we obtain $R_4 = O_p(\|\hat{\mu} - \mu_0\|_\infty \sigma_n/\sqrt{n}) = o_p(\sigma_n/\sqrt{n})$ by Assumption 5(b).

Averaging over the $\mathcal{K}$ folds yields

$$\tilde{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^n v_{K_n}^*(X_i, D_i) \epsilon_i + o_p(\sigma_n/\sqrt{n}).$$

The summands $v_{K_n}^*(X_i, D_i) \epsilon_i / \sigma_n$ form a triangular array of i.i.d. random variables with mean zero (since $\mathbb{E}[\epsilon_i | X_i, D_i] = 0$) and unit variance (by the definition of $\|\cdot\|_{sd}$); the Lindeberg condition is verified under Assumption 5(a) exactly as in the proof of Proposition 1 in Chen and Gao (2026). Asymptotic normality of $\sqrt{n}(\tilde{\theta} - \theta_0) / \sigma_n$ then follows from the Lindeberg–Feller central limit theorem, and the studentized statement from Slutsky’s theorem whenever $\hat{\sigma}_n^2 / \sigma_n^2 \xrightarrow{p} 1$. $\square$

Proof of Corollary 2. Take $\Theta = \bar{V}$. By Theorem 3, the pathwise derivative $D_\mu \bar{V}(\mu_0)[\cdot]$ is a submanifold integral of dimension $m = d_x - 1$ and $\sigma_n^2 = \|v_{K_n}^*\|_{sd}^2 \asymp K_n^{1/d_x}$, so $\sigma_n / \sqrt{n} \asymp \sqrt{K_n^{1/d_x} / n}$ and Assumption 5(c)–(e) reduce to the value-functional rates discussed in Section 5.6. The conclusion is immediate from Theorem 7, with consistency of the sieve variance estimator $\hat{\sigma}_{\bar{V}}^2$ supplied by Section 4.1. $\square$

Proof of Corollary 1. Take $\Theta = \bar{W}$. By the generalized Leibniz rule of Section 2.3 the boundary contribution to $D_\mu \bar{W}(\mu_0)[\cdot]$ vanishes, and by Theorem 1 the derivative admits the bounded global Riesz representer $\nu^* \in \mathcal{V}$. Hence $v_{K_n}^* = \Pi_{K_n} \nu^*$ converges to ν^* in $\|\cdot\|$, and $\sigma_n = \|v_{K_n}^*\|_{sd} \rightarrow \|\nu^*\|_{sd} = \sigma_W < \infty$; thus $\sigma_n / \sqrt{n} \asymp n^{-1/2}$ and Assumption 5(c) coincides with the $o_p(n^{-1/2})$ product rate of Remark 4. Theorem 7 then gives $\sqrt{n}(\tilde{\theta} - W(h_0)) \xrightarrow{d} \mathcal{N}(0, \sigma_W^2)$ in the known-density case. In the unknown-density case the deterministic integral $\bar{W}(\hat{\mu}_{-\kappa})$ in (24) is replaced by the sample-average welfare term, which contributes the additional variance $\text{Var}([h_0]_+)$, yielding the limiting variance $\bar{\sigma}_W^2$ of Theorem 2. The asymptotically equivalent closed-form (AIPW) construction of Remark 4 is proved separately in Appendix A. $\square$

C Additional Simulation Results

C.1 Theorem 1

In principle, a variety of nonparametric estimators can be employed in the first step for the nuisance parameters, provided their convergence rates are sufficiently fast. In this subsection, we assess the estimation and inference performance of the welfare functional estimator under a known target distribution F , using a generalized additive model (GAM) as the first-stage estimator. Specifically, for Models 1–3, we use B-splines as the smooth terms in GAM, while for Models 4–7, we adopt the default tensor product smooths with cubic regression splines.

The simulation results are presented in Table 11. In Models 6 and 7, the confidence intervals exhibit overcoverage relative to the nominal 95% level. This likely reflects the slow convergence of the variance plug-in estimator toward the true asymptotic variance of

the welfare estimator, potentially due to suboptimal choices of smooth terms in the GAM specification.

Table 11: Theorem 1 Simulation Results for Models 1–7 (GAM)

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M1	1500	0.3857	0.0062	0.0474	0.0481	0.0035	0.9500
	3000	0.3857	0.0025	0.0341	0.0339	0.0018	0.9525
	6000	0.3857	-0.0009	0.0231	0.0239	0.0009	0.9545
M2	1500	0.2358	-0.0092	0.0445	0.0513	0.0027	0.9670
	3000	0.2358	-0.0057	0.0335	0.0365	0.0013	0.9610
	6000	0.2358	-0.0048	0.0239	0.0259	0.0006	0.9615
M3	1500	0.5001	0.0037	0.0587	0.0604	0.0022	0.9580
	3000	0.5001	0.0021	0.0409	0.0430	0.0011	0.9605
	6000	0.5001	-0.0002	0.0294	0.0305	0.0006	0.9600
M4	1500	0.1033	0.0303	0.0465	0.0550	0.0065	0.9615
	3000	0.1033	0.0193	0.0338	0.0394	0.0036	0.9635
	6000	0.1033	0.0130	0.0243	0.0281	0.0021	0.9610
M5	1500	0.0499	0.0316	0.0407	0.0527	0.0087	0.9685
	3000	0.0499	0.0217	0.0294	0.0372	0.0050	0.9680
	6000	0.0499	0.0138	0.0218	0.0261	0.0030	0.9635
M6	1500	0.2315	0.0035	0.0450	0.0634	0.0039	0.9930
	3000	0.2315	0.0004	0.0314	0.0448	0.0019	0.9965
	6000	0.2315	-0.0008	0.0234	0.0317	0.0010	0.9920
M7	1500	0.1250	0.0040	0.0380	0.0545	0.0064	0.9950
	3000	0.1250	0.0026	0.0272	0.0385	0.0030	0.9920
	6000	0.1250	0.0016	0.0199	0.0272	0.0015	0.9915

Notes: (1) W_{true} denotes the true welfare functional, computed numerically using 5,000 Sobol points drawn from F . (2) Bias is the average deviation from W_{true} . (3) SD is the sampling standard deviation. (4) SE is the average estimated asymptotic standard error of $\hat{W}(\hat{\mu})$. (5) SD(SE) is the standard deviation of SE across iterations. (6) Coverage is the empirical coverage probability of the 95% CI.

C.2 Theorem 2

We next examine the estimation and inference performance of the welfare functional estimator under an unknown target distribution F , using GAM as the first-stage estimator. The GAM specifications are identical to those described in the previous subsection. The simulation results, reported in Table 12, demonstrate that our proposed inference procedure

exhibits strong finite-sample performance.

Table 12: Theorem 2 Simulation Results for Models 8–14 (GAM)

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M8	1,500	0.3857	0.0077	0.0412	0.0421	0.0025	0.951
	3,000	0.3857	0.0036	0.0296	0.0297	0.0013	0.952
	6,000	0.3857	0.0008	0.0206	0.0209	0.0007	0.956
M9	1,500	0.2358	0.0000	0.0414	0.0436	0.0021	0.965
	3,000	0.2358	0.0008	0.0297	0.0310	0.0010	0.962
	6,000	0.2358	0.0002	0.0206	0.0220	0.0005	0.960
M10	1,500	0.5001	0.0059	0.0512	0.0514	0.0018	0.952
	3,000	0.5001	0.0033	0.0357	0.0366	0.0010	0.957
	6,000	0.5001	0.0010	0.0252	0.0260	0.0005	0.962
M11	1,500	0.1033	0.0266	0.0362	0.0396	0.0036	0.938
	3,000	0.1033	0.0154	0.0261	0.0284	0.0020	0.940
	6,000	0.1033	0.0094	0.0190	0.0202	0.0011	0.942
M12	1,500	0.0499	0.0277	0.0324	0.0370	0.0052	0.941
	3,000	0.0499	0.0172	0.0232	0.0263	0.0029	0.946
	6,000	0.0499	0.0106	0.0169	0.0187	0.0016	0.941
M13	1,500	0.2315	0.0062	0.0431	0.0456	0.0024	0.964
	3,000	0.2315	0.0027	0.0308	0.0323	0.0012	0.959
	6,000	0.2315	0.0007	0.0218	0.0229	0.0006	0.957
M14	1,500	0.1250	0.0056	0.0367	0.0394	0.0043	0.957
	3,000	0.1250	0.0023	0.0249	0.0276	0.0019	0.968
	6,000	0.1250	0.0007	0.0176	0.0195	0.0009	0.969

Notes: (1) W_{true} denotes the true welfare functional, computed numerically using 5,000 Sobol points drawn from F . (2) Bias is the average deviation from W_{true} . (3) SD is the sampling standard deviation. (4) SE is the average estimated asymptotic standard error of $\hat{W}(\hat{\mu})$. (5) SD(SE) is the standard deviation of SE across iterations. (6) Coverage is the empirical coverage probability of the 95% CI.

C.3 Theorem 3

In addition to the model specification considered in the main text for the Theorem 3 simulations, we conducted two auxiliary simulation studies with slightly modified designs: one with the target population $F \sim \text{U}([-1.75, 1.75]^2)$ and the other with $F \sim \text{U}([-1.9, 1.9]^2)$, keeping all other settings fixed. The results are reported in Table 13. We find that the coverage rate approaches 95% as the sample size increases in both cases. However, in the latter case the supports of the treated and control groups do not fully nest the target pop-

ulation F , potentially leading to extrapolation bias of the first-stage estimator and a slight undercoverage.

Table 13: Sensitivity Checks for Model 15: Target Support

Case	n	V_{true}	Bias	SD	SE	Coverage
(a) $F \sim \text{U}[-1.75, 1.75]^2$	1500	3.1416	0.0222	0.1289	0.1198	0.926
	3000	3.1416	0.0077	0.0801	0.0796	0.942
	6000	3.1416	0.0089	0.0542	0.0558	0.952
(b) $F \sim \text{U}[-1.9, 1.9]^2$	1500	3.1416	0.0118	0.0746	0.0676	0.923
	3000	3.1416	0.0047	0.0474	0.0453	0.938
	6000	3.1416	0.0037	0.0334	0.0317	0.936

Besides the sieve dimensions for estimating $\mu_0(x, 1)$ and $\mu_0(x, 0)$, ϵ is an additional tuning parameter that determines how well the level set is approximated. To assess the sensitivity of our method to this parameter, we consider the alternative values $\epsilon = 0.0025$ and $\epsilon = 0.0075$ while keeping everything else fixed. Table 14 shows that the simulation results are not materially affected by the choice of ϵ .

Table 14: Sensitivity Checks for Model 15: Tuning Parameter ϵ

Case	n	V_{true}	Bias	SD	SE	Coverage
(a) $\epsilon = 0.0075$	1500	3.1416	0.0120	0.0642	0.0635	0.9395
	3000	3.1416	0.0126	0.0442	0.0448	0.9445
	6000	3.1416	0.0102	0.0311	0.0316	0.9445
(b) $\epsilon = 0.0025$	1500	3.1416	0.0120	0.0642	0.0638	0.9420
	3000	3.1416	0.0126	0.0442	0.0450	0.9445
	6000	3.1416	0.0102	0.0311	0.0317	0.9450

C.4 Simulation Results for DML

An alternative to the semi-parametric two-step estimation of the welfare functional is to use the double/debiased machine learning (DML) approach of [Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey and Robins \(2018\)](#). The DML estimator is implemented via a cross-fitting scheme. Specifically, the sample is partitioned into $\mathcal{K}$ folds. For each fold, we use the observations in the remaining $\mathcal{K} - 1$ folds to obtain nonparametric estimates of the nuisance functions $\mu_0(x, 1)$ and $\mu_0(x, 0)$ (in our case, using GAM). These estimates are then applied to the held-out fold to take an average of $[\hat{\mu}(X_i, 1) - \hat{\mu}(X_i, 0)]_+$. This process is repeated so that each fold serves once as the testing set, and the final estimator

is obtained by averaging the resulting values across all folds. A confidence interval could then be constructed as in Theorem 2 with the debiased ML estimate in place of the semi-parametric two-step estimate.

We simulate 1,000 iterations for Models 1 to 3 with $\mathcal{K} = 5$, and the results are summarized in Table 15. The confidence intervals exhibit slight undercoverage in small samples, but this issue diminishes as the sample size increases. This pattern is consistent with the asymptotic nature of the interval’s construction.

Table 15: DML Simulation Results for the Welfare Functional

Model	n	W_{true}	Bias	SD	SE	Coverage
Model 1	1,500	0.3857	-0.01271	0.04583	0.04215	0.906
Model 1	3,000	0.3857	-0.00776	0.03184	0.02969	0.921
Model 1	6,000	0.3857	-0.00392	0.02193	0.02092	0.934
Model 2	1,500	0.2358	-0.01051	0.04900	0.04372	0.916
Model 2	3,000	0.2358	-0.00475	0.03319	0.03097	0.934
Model 2	6,000	0.2358	-0.00195	0.02297	0.02200	0.929
Model 3	1,500	0.5001	-0.00568	0.05636	0.05134	0.913
Model 3	3,000	0.5001	-0.00261	0.03701	0.03654	0.953
Model 3	6,000	0.5001	-0.00174	0.02628	0.02597	0.954

C.5 Sieve Variance Estimation

Table 16: Theorem 1 Simulation Results for Models 1–3 (Sieve Variance)

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M1	1500	0.3857	0.0152	0.0469	0.0467	0.0035	0.9370
	3000	0.3857	0.0076	0.0337	0.0329	0.0019	0.9390
	6000	0.3857	0.0036	0.0229	0.0232	0.0011	0.9510
M2	1500	0.2358	0.0039	0.0481	0.0488	0.0029	0.9480
	3000	0.2358	0.0042	0.0350	0.0345	0.0014	0.9425
	6000	0.2358	0.0025	0.0241	0.0245	0.0007	0.9535
M3	1500	0.5001	0.0038	0.0580	0.0581	0.0023	0.9560
	3000	0.5001	0.0019	0.0407	0.0413	0.0012	0.9560
	6000	0.5001	-0.0002	0.0293	0.0294	0.0006	0.9525

Table 17: Theorem 1 Simulation Results for Models 4–7 (Sieve Variance)

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M4	1500	0.1033	0.0185	0.0478	0.0482	0.0088	0.9400
	3000	0.1033	0.0088	0.0348	0.0350	0.0048	0.9445
	6000	0.1033	0.0043	0.0248	0.0251	0.0024	0.9490
M5	1500	0.0499	0.0282	0.0418	0.0431	0.0110	0.9310
	3000	0.0499	0.0158	0.0303	0.0311	0.0068	0.9335
	6000	0.0499	0.0094	0.0227	0.0224	0.0038	0.9290
M6	1500	0.2315	0.0268	0.0559	0.0552	0.0057	0.9260
	3000	0.2315	0.0123	0.0394	0.0402	0.0030	0.9480
	6000	0.2315	0.0050	0.0288	0.0288	0.0015	0.9490
M7	1500	0.1250	0.0462	0.0505	0.0502	0.0087	0.8920
	3000	0.1250	0.0218	0.0346	0.0350	0.0046	0.9270
	6000	0.1250	0.0101	0.0245	0.0245	0.0023	0.9400

Table 18: Theorem 2 Simulation Results for Models 8–10 (Sieve Variance)

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M8	1500	0.3857	0.0068	0.0414	0.0418	0.0026	0.9475
	3000	0.3857	0.0029	0.0296	0.0294	0.0014	0.9500
	6000	0.3857	0.0006	0.0206	0.0208	0.0007	0.9550
M9	1500	0.2358	0.0042	0.0425	0.0432	0.0026	0.9570
	3000	0.2358	0.0029	0.0304	0.0305	0.0012	0.9515
	6000	0.2358	0.0012	0.0209	0.0216	0.0006	0.9505
M10	1500	0.5001	0.0067	0.0511	0.0512	0.0020	0.9480
	3000	0.5001	0.0037	0.0357	0.0364	0.0011	0.9595
	6000	0.5001	0.0013	0.0253	0.0259	0.0006	0.9580

Table 19: Theorem 2 Simulation Results for Models 11–14 (Sieve Variance)

Model	n	W_{true}	Bias	SD	SE	SD(SE)	Coverage
M11	1500	0.1033	0.0307	0.0365	0.0379	0.0042	0.9070
	3000	0.1033	0.0156	0.0264	0.0272	0.0023	0.9285
	6000	0.1033	0.0084	0.0192	0.0195	0.0013	0.9380
M12	1500	0.0499	0.0418	0.0316	0.0345	0.0050	0.8400
	3000	0.0499	0.0232	0.0229	0.0246	0.0031	0.8930
	6000	0.0499	0.0126	0.0168	0.0175	0.0018	0.9185
M13	1500	0.2315	0.0177	0.0432	0.0437	0.0031	0.9415
	3000	0.2315	0.0088	0.0309	0.0311	0.0016	0.9435
	6000	0.2315	0.0041	0.0221	0.0221	0.0008	0.9475
M14	1500	0.1250	0.0251	0.0382	0.0387	0.0051	0.9200
	3000	0.1250	0.0117	0.0258	0.0268	0.0024	0.9430
	6000	0.1250	0.0054	0.0182	0.0188	0.0012	0.9425

D Additional Results for Empirical Application

D.1 Distribution of the Outcome Variable

The wide welfare-gain confidence intervals, and the share confidence intervals whose upper endpoints exceed one in Table 10, are driven by the distribution of the 30-month earnings outcome rather than by the inference procedure. Figure 1 plots its histogram. The outcome is heavily right-skewed: across the JTPA sample ($n = 9,223$) its mean is \$16,093 against a median of \$11,187, its standard deviation (\$17,071) exceeds its mean, its maximum (\$155,760) is about ten times the mean, and 11.8% of individuals report exactly zero earnings; the (Fisher) skewness is 1.55 and the excess kurtosis is 3.53. The \$774 per-treatment cost considered in the second specification amounts to only about 4.5% of one outcome standard deviation. We use the Kitagawa and Tetenov (2018) replication package without any further transformation of the outcome, so that our point estimates remain comparable to theirs; the price is that the large outcome variance inflates the estimated standard errors and hence the interval widths.

D.2 Sensitivity Analysis for Tuning Parameters

When conducting inference for the value functional in the empirical analysis of the JTPA data, two tuning parameters must be specified: ι and s . The parameter ι determines the size of the set $\{x \in \mathcal{X} : -\epsilon < \hat{h}(x) < \epsilon\}$. The parameter s controls the smoothness of

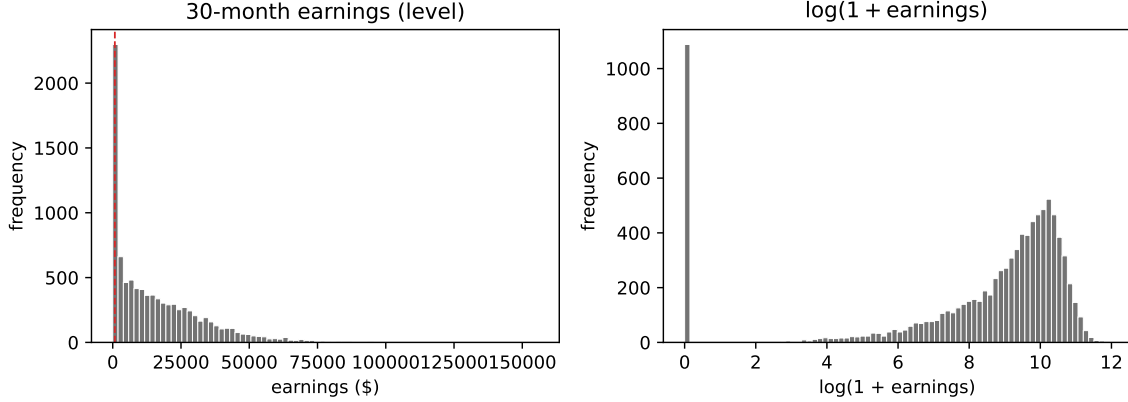


Figure 1: Distribution of the JTPA 30-month earnings outcome ($n = 9,223$). Left: levels, with the dashed line marking the \$774 per-treatment cost; the distribution is heavily right-skewed, with a point mass at zero and extreme upper outliers (maximum \$155,760). Right: $\log(1 + \text{earnings})$.

the estimated density function over the trimmed dataset by scaling the bandwidths used in kernel density estimation.

To evaluate robustness, we perform sensitivity analyses with respect to both tuning parameters. The results, reported in Table 20, show that the estimated value functional $\hat{V}(\hat{\mu})$ is largely insensitive to the choice of ι , but more responsive to the choice of s . This highlights the importance of choosing an appropriate kernel density estimator.

D.3 Sensitivity Analysis for Density Estimation

We employed a naive Gaussian kernel density estimator for the covariate distribution in the trimmed dataset, treating years of education as a continuous variable. While years of education is theoretically continuous, in practice it takes on only discrete values in the dataset. This motivates considering alternative density estimators that explicitly treat years of education as categorical in order to assess the robustness of our inference. To this end, we partitioned the trimmed dataset by educational level and examined three cases: (1) a naive Gaussian kernel density estimator with bandwidths selected by `Hscv()` and scaled by a small factor to ensure smoothness; (2) a logspline density estimator (Stone, Hansen, Kooperberg and Truong, 1997); (3) a penalized B-spline density estimator (Schellhase and Kauermann, 2012).

The results are presented in Table 21. As shown, when costs are taken into account, the Gaussian kernel density estimator discussed in the main text yields relatively conservative confidence intervals compared to the three alternatives. In contrast, when costs are not considered, the results remain largely unchanged across all the estimators. However, we

Table 20: Sensitivity of $\hat{V}(\hat{\mu})$ to Tuning Parameters ι and s (Cost = F vs. Cost = T)

(a) Sensitivity to ι					
ι	$\hat{V}$	SE	CI Low	CI High	Num
Cost = F					
0.005	0.8908	0.0866	0.7210	1.0606	475
0.0075	0.8908	0.0862	0.7219	1.0597	713
0.0100	0.8908	0.0828	0.7286	1.0530	931
0.0125	0.8908	0.0840	0.7262	1.0554	1203
0.0150	0.8908	0.0857	0.7228	1.0588	1463
Cost = T					
0.005	0.7971	0.1343	0.5338	1.0603	689
0.0075	0.7971	0.1303	0.5417	1.0525	1044
0.0100	0.7971	0.1373	0.5279	1.0662	1411
0.0125	0.7971	0.1399	0.5229	1.0712	1784
0.0150	0.7971	0.1406	0.5214	1.0727	2152

(b) Sensitivity to s					
s	$\hat{V}$	SE	CI Low	CI High	Num
Cost = F					
1	0.8908	0.0896	0.7151	1.0665	931
2	0.8908	0.0860	0.7224	1.0593	931
3	0.8908	0.0828	0.7286	1.0530	931
4	0.8908	0.0809	0.7322	1.0494	931
5	0.8908	0.0793	0.7354	1.0462	931
Cost = T					
1	0.7971	0.1103	0.5809	1.0132	1411
2	0.7971	0.1254	0.5513	1.0428	1411
3	0.7971	0.1373	0.5279	1.0662	1411
4	0.7971	0.1420	0.5187	1.0754	1411
5	0.7971	0.1554	0.4925	1.1016	1411

Notes: (a) Cost = T if the outcome variable is the 30-month earnings minus an additional \$774 and F otherwise. (b) $SE = \hat{\sigma}_V / \sqrt{n}$. (c) $\epsilon = \iota \times SD(\hat{h}(x))$ over **X__trim**. (d) $CI\ Low = \hat{V}(\hat{\mu}) - 1.96 \times SE$, $CI\ High = \hat{V}(\hat{\mu}) + 1.96 \times SE$. (e) Num is the number of Sobol points whose $\hat{h}$ evaluations fall into $\{x \in \mathbf{X_trim} : |\hat{h}(x)| < \epsilon\}$.

should emphasize that these three alternative density estimators are applied based on the assumption that years of education is categorical, but in theory it is treated as a continuous variable, which provides justification for using the Gaussian kernel estimator in the main text.

Table 21: Sensitivity of $\hat{V}(\hat{\mu})$ to Alternative Density Estimators

Estimator	Cost = T				Cost = F			
	$\hat{V}$	SE	CI Low	CI High	$\hat{V}$	SE	CI Low	CI High
Logspline	0.7971	0.0932	0.6144	0.9797	0.8908	0.1041	0.6868	1.0948
Kernel	0.7971	0.0704	0.6590	0.9351	0.8908	0.0980	0.6988	1.0829
Penalized B-spline	0.7971	0.0601	0.6793	0.9148	0.8908	0.0691	0.7555	1.0262

Notes: (a) Cost = T if the outcome variable is 30-month earnings minus an additional \$774, and Cost = F otherwise. (b) $SE = \hat{\sigma}_V/\sqrt{n}$. (c) $CI\ Low = \hat{V}(\hat{\mu}) - 1.96 \times SE$, $CI\ High = \hat{V}(\hat{\mu}) + 1.96 \times SE$.

D.4 Extrapolating Sieve Estimates

We also examine the case where the dataset is not trimmed. In this setting, interpolation is required to evaluate the estimated functions $\hat{\mu}(x, 0)$ and $\hat{\mu}(x, 1)$ outside the supports of the control and treated groups, respectively. The formulas for the welfare and value functional estimators, as well as their asymptotic variances, remain unchanged. The only difference is that the sample averages are now computed over the full covariate support rather than the trimmed dataset. The results for this specification are reported in Table 22.

D.5 Estimating $\hat{\Omega}$ Using the Trimmed Dataset

We used the full sample to estimate the asymptotic variance-covariance matrix $\hat{\Omega}$. For comparison, Table 23 reports the results obtained when $\hat{\Omega}$ is instead estimated using only the trimmed dataset. We observe that the CIs expand by a small factor.

D.6 The Sieve Score Bootstrapping Procedure for Critical Values

Following Chen and Christensen (2018), one could also use the sieve score bootstrap procedure to calculate the critical value used in the construction of the 95% CI for $\bar{V}(\mu_0)$. The algorithm starts with making iid draws $\{\omega_i\}_{i=1}^n$ from a distribution independent of the data with mean zero, unit variance, and finite third moment (e.g. a standard normal distribution),

Table 22: Estimated Welfare Gains and Share of Population to be Treated Under Nonparametric Plug-in Rule Without Trimming

(a) 30-Month Post-Program Earnings, No Treatment Cost		
Method	Share Treated	Est. Welfare Gain
Ours	0.92 (0.80, 1.03)	\$1,459 (\$825, \$2,093)
KT (2018)	0.91 NA	\$1,693 NA
(b) 30-Month Post-Program Earnings, \$774 Cost per Treatment		
Method	Share Treated	Est. Welfare Gain
Ours	0.85 (0.71, 0.99)	\$768 (\$164, \$1,373)
KT (2018)	0.78 NA	\$996 NA

Note: Two-sided 95% confidence intervals in parentheses, constructed based on the asymptotic distributions of the corresponding estimators.

Table 23: Estimated Share of Population to be Treated Under Nonparametric Plug-in Rule With $\hat{\Omega}$ Estimated Using Trimmed Data

(a) 30-Month Post-Program Earnings, No Treatment Cost	
Method	Share Treated
Ours	0.89 (0.72, 1.06)
KT (2018)	0.91 NA
(b) 30-Month Post-Program Earnings, \$774 Cost per Treatment	
Method	Share Treated
Ours	0.80 (0.44, 1.15)
KT (2018)	0.78 NA

and then calculates the bootstrap sieve t-statistic using the formula

$$Z_n^* = \frac{1}{\hat{\sigma}_V} \hat{D}_\mu \bar{V}(\hat{\mu}) [\bar{\psi}]' (B'B/n)^{-1} \begin{bmatrix} \frac{1}{\sqrt{n}} \sum_{i=1}^{n_1} \psi^{(K_1)}(X_i) \hat{u}_i \omega_i \\ \frac{1}{\sqrt{n}} \sum_{j=n_1+1}^n \psi^{(K_0)}(X_j) \hat{u}_j \omega_j \end{bmatrix}$$

with B being the design matrix of the regression of Y_i on $D_i \psi^{(K_1)}(X_i)$ and $(1 - D_i) \psi^{(K_0)}(X_i)$, and n_1 the number of treated observations (with the data ordered so that the treated observations come first). The 95% quantile of the bootstrapped $|Z_n^*|$ could be used as the critical value in the construction of the CI. Setting the number of bootstrap draws to 1,000, the resulting critical value is 1.866, which is not significantly different from 1.96.